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Minimum Wages in the 21st Century

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Introduction



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- Minimum wage is popular in the general population and among policy makers
- Today more than 90% of the countries have some form of labour regulation on wages (ILO, 2017)
- Many countries apply nation-wide minimum wages covering all industries, and almost all workers
- The level of minimum wage is at a historically high-level and strong political movements to push it further



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Section 1: A brief history of minimum wages

Why do we have minimum wages?



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The motivation beyond minimum wage regulations varies across space and time

- **First adopters** (New Zealand in 1894; Australian state Victoria in 1896, UK in 1909): aimed to use this as a policy tool to avoid race-to-the-bottom competition

First adopters



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- First adopters (New Zealand in 1894; Australian state Victoria in 1896, UK in 1909) aimed to use this as a policy tool to avoid race to the bottom competition
- Key goal is to stop the proliferation of “sweatshops” in 1890s (Nordlund, 1997)
- Sweatshops mainly employed women and young workers and paid them substandard wages
 - Many efficient and worthwhile companies employing working class breadwinners lost market share
 - The minimum wage sought to reverse these trends
 - Minimum Wages were often industry-specific

USA



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- US states started to pass minimum wage laws in 1912
- 17 states passed some sort of wage regulation by 1923
- Federal minimum wage was introduced in 1938
- The main goal was to
 - support the safety and well-being of workers
 - protect interstate commerce and avoid race-to-the-bottom competition (Newell 2009)
- Restricted to specific industries, occupations
- Over time the the minimum wage was expanded to more industries and occupations (see Brown, 1999, Montialoux and Derenoncourt, 2021)

Latin America



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- During 1930s-1940s Latin America also saw a big expansion in the adoption of minimum wage laws (Grimshaw and Miozzo 2003)
- A common aspect: the countries' constitution recognised the right of workers to obtain wages sufficient for a decent standard of living
- Political economy considerations:
 - State intervention in labour relations was also motivated by an effort to make workers look at the state instead of unions to protect their interests
 - Unions were often linked to radical political ideologies questioning state's authority
- Similar considerations in Nordic countries where strong unions opposing the minimum wage

Europe, Africa, Asia



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- Many Western Europe started adopting minimum wage policies after the Second World War (Luxembourg, 1945; France, 1950, Spain, 1963; Netherlands 1969)
 - Motivated by workers' right to be protected against extremely low wages (ILO 2017)
- Africa followed as their regulations were tightly linked to the colonial powers in Europe
- Asia was relatively slower, not seeing an important increase in adoption until recent decades

Today minimum wages are often motivated by concerns about inequality

Why do we have minimum wages?



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The motivation beyond minimum wage regulations varies across space and time

- **First adopters** (New Zealand in 1894; Australian state Victoria in 1896, UK in 1909): aimed to use this as a policy tool to avoid race-to-the-bottom competition
- **United States** (Massachusetts in 1912, then federally in 1938): support the safety and well-being of workers, protect interstate commerce, and avoid race-to-the-bottom competition (Newell 2009)
- **Latin America** (1930s and 1940s): strong rights for workers in the constitutions and state intervention in labour relations were used to look at the state instead of unions to protect their interests
- **Western Europe** (post WWII), part of the welfare state and worker protection. Some historically relied on collective bargaining, but decline in coverage led to national minimum wage (e.g., Germany).
- **Africa** (post WWII): regulations were tightly linked to the colonial powers in Europe
- **Asia** (post WWII and 90s-2000s): first wave followed the post war recovery and associated with introducing ILO standards (India, Japan, Taiwan). The second wave is often associated with economic deregulations (e.g. in case of South Korea, Indonesia, China, Vietnam)



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Section 2: The Minimum Wage Debate

The Minimum Wage Debate



- Long-standing tradition arguing that a minimum wage will reduce employment if it is set above the competitive level (Mill, 1848; Marshall, 1897; Stigler, 1946)
- These predictions have been formalized in the neoclassical framework (Hicks, 1932)
- Key assumption (or so-called “price theory”)
 - Low-skilled workers are homogenous
 - Many firms compete for the same type of workers
 - Low-skilled jobs are standardized

“Neoclassical” Firm’s Problem



- Firms maximize their profits, taking market-level wage as given:

$$\max_{L_j} pf(L_j) - wL_j$$

- The FOC:

$$pf'(L_j) - w = 0$$

or

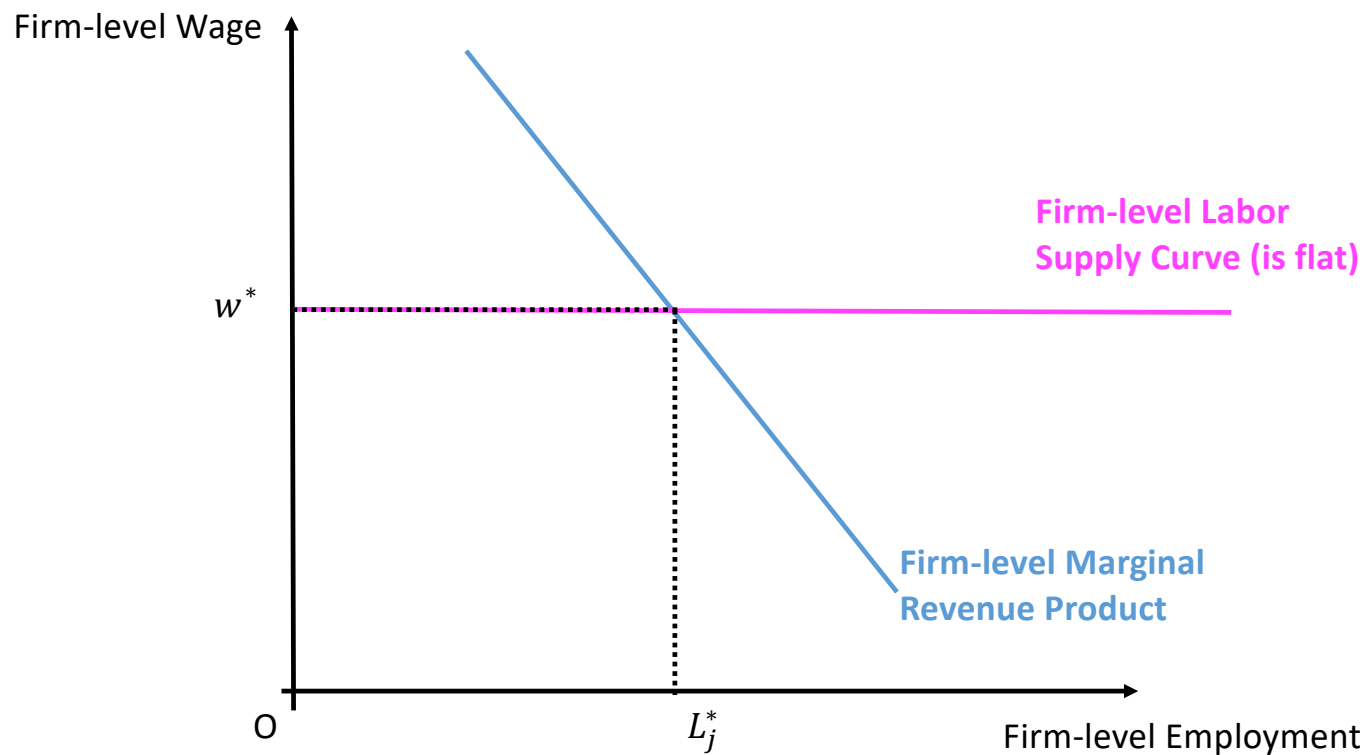
$$\underbrace{pf'(L_j)}_{\text{Marginal Revenue Product}} = \underbrace{w}_{\text{Marginal Cost}}$$

$$\text{And so } L_j = f'^{-1}\left(\frac{w}{p}\right)$$

“Neoclassical” Firm’s Problem



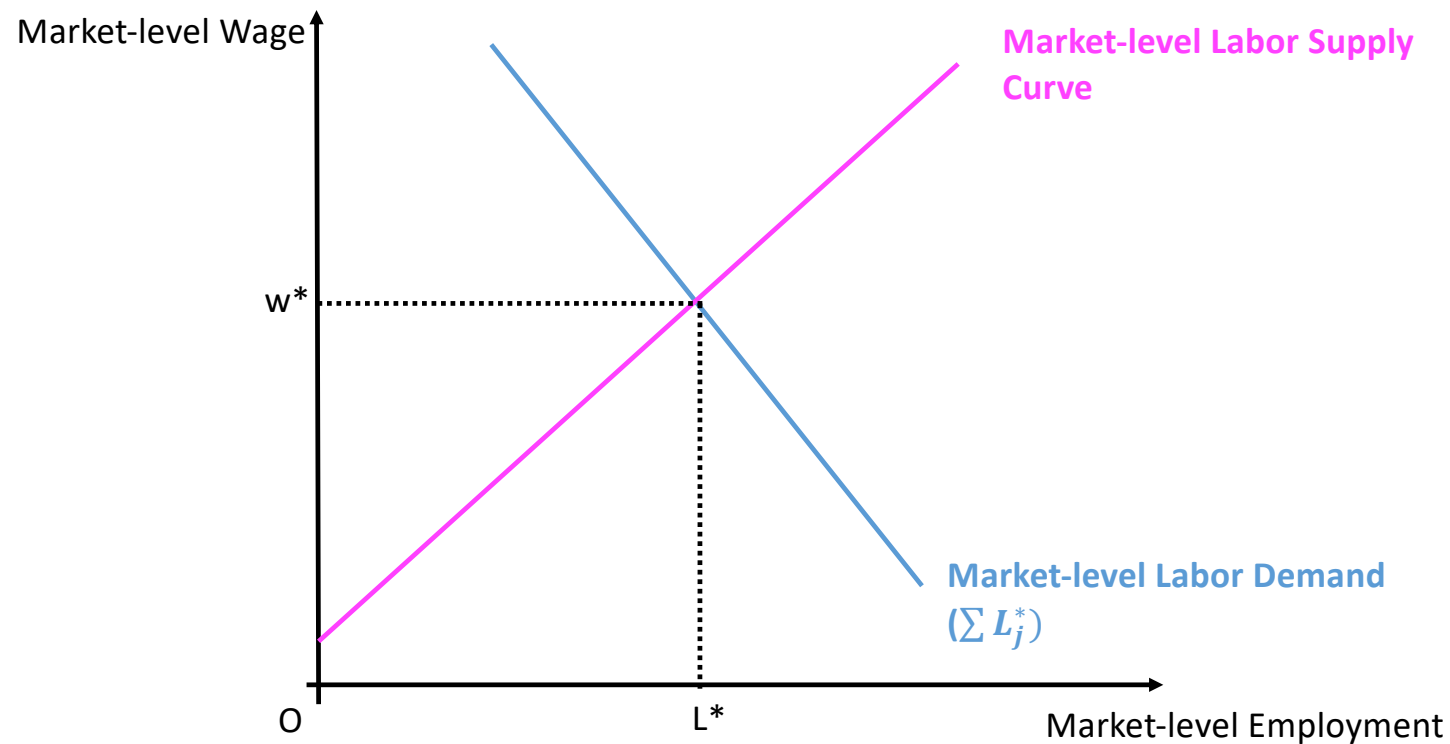
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Market-level Equilibrium



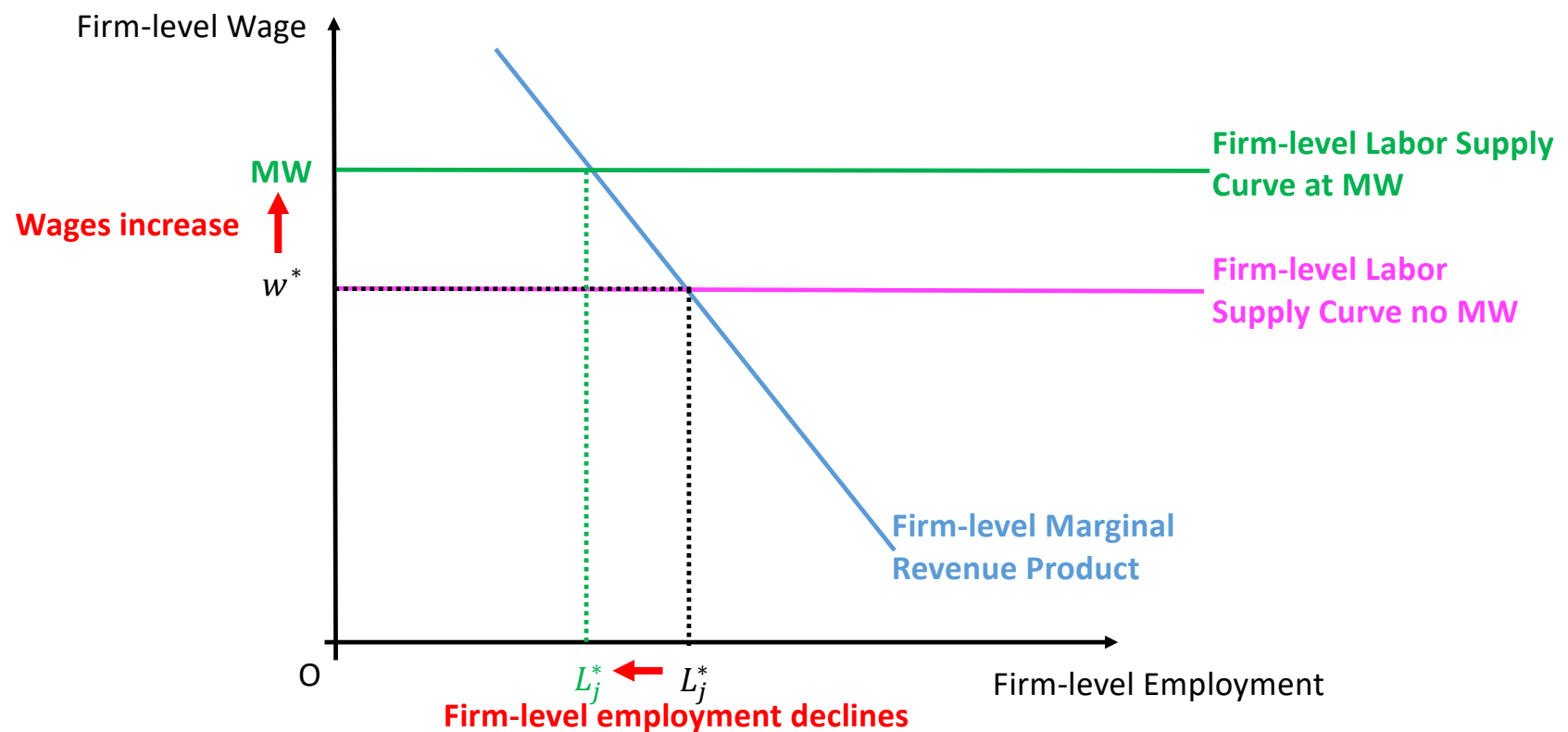
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The Effect of the Minimum Wage



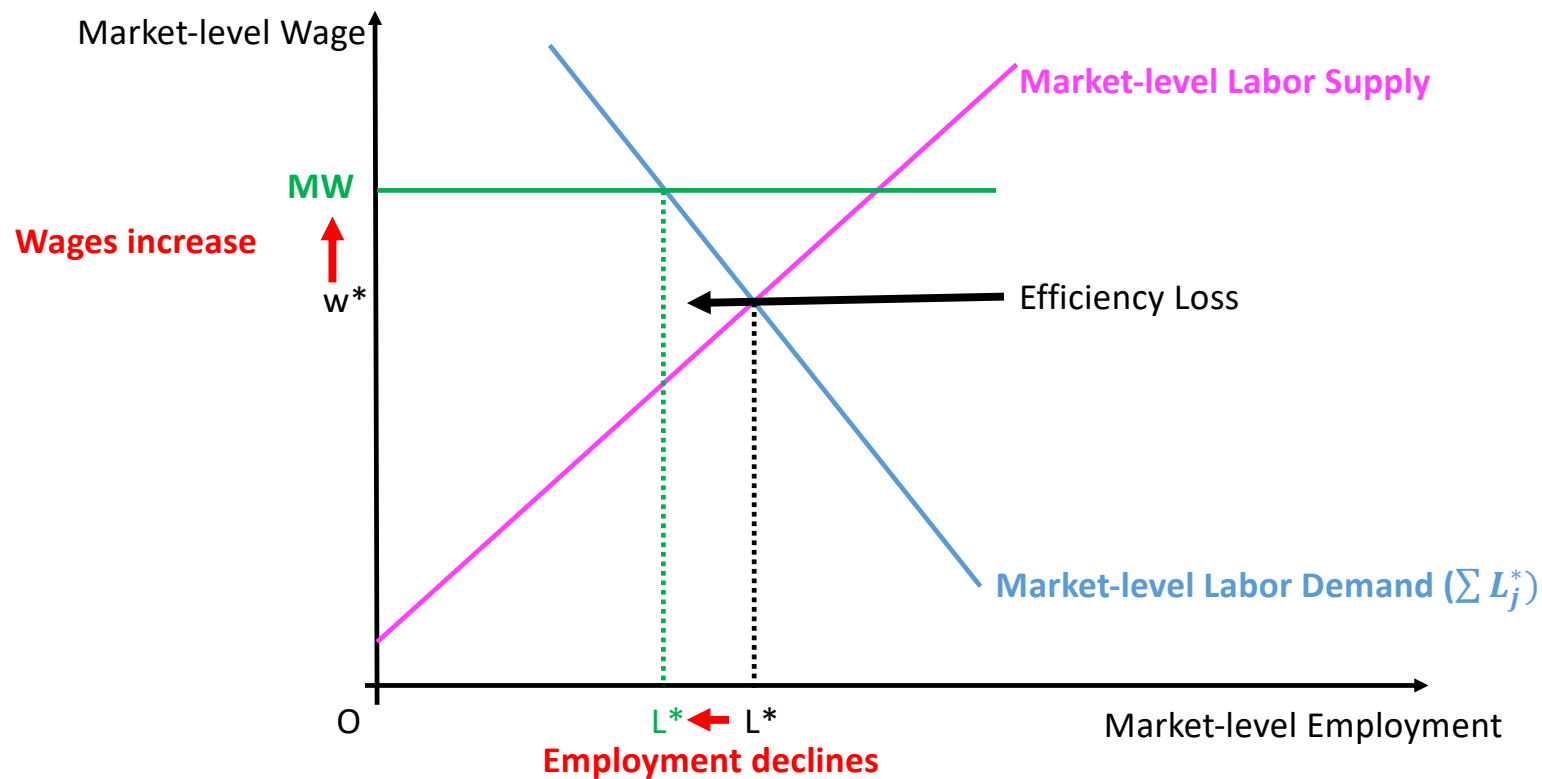
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The Effect of Minimum Wages on Market-level Outcomes



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The Minimum Wage Debate



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- Long-standing tradition arguing that a minimum wage will reduce employment if it is set above the competitive level (Mill, 1848; Marshall, 1897; Stigler, 1946)
- These predictions have been formalized in the neoclassical framework (Hicks, 1932)
- Key assumption (or so-called “price theory”)
 - Low-skilled workers are homogenous
 - Many firms compete for the same type of workers
 - Low-skilled jobs are standardized

Key Insight: Minimum wages have an unambiguously negative effect on employment!

For many years, the policy served as an exemplar of how interference in market equilibrium could do more harm than good—even with the best of intentions (Sidgwick, 1886; Marshall, 1897, Stigler, 1946)

The Minimum Wage Debate



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George Stigler (1946): price theory

- No free lunch
- Minimum Wage decreases employment and leads to efficiency loss
- This is a clear testable prediction of the theory

Richard Lester (1947): common sense approach

- Instead of analysing models, look at the data
- Ask managers about what determines employment
- Most important factor: the output demand
- Less than 20% reported the labour cost as the most important
- **Conclusion:** (small) increase of labour cost will have a limited employment effect

The Consensus



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Milton Friedman (1953): “Essays in Positive Economics”

- Assumptions can be unrealistic and simplified

The billiard player analogy:

*Consider the problem of predicting the shots made by an expert billiard player. It seems not at all unreasonable that **excellent predictions would be yielded** by the hypothesis that the billiard player made his shots as if he knew **the complicated mathematical formulas that would give the optimum directions of travel**, could estimate accurately by eye the angles, etc., describing the location of the balls, could make lightning calculations from the formulas, and could then make the balls travel in the direction indicated by the formulas.*

*Our confidence in **this hypothesis is not based on the belief that billiard players, even expert ones, can or do go through the process described**; it derives rather from the belief that, unless in some way or other they were capable of reaching essentially the same result, they would not in fact be expert billiard players.*

The Consensus



Milton Friedman (1953): “Essays in Positive Economics”

- Assumptions can be unrealistic and simplified
- A good model focuses on the most important mechanisms, instead of discussing all irrelevant ingredients
- A good model can explain and predict what happens if circumstances change
- Evidence produced over the ‘60s-80s supports declining employment
- Neoclassical model/price theory can capture what happens in response to the minimum wage

Consensus Breaks



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Card and Krueger (1994, 1995) and the “credibility revolution”

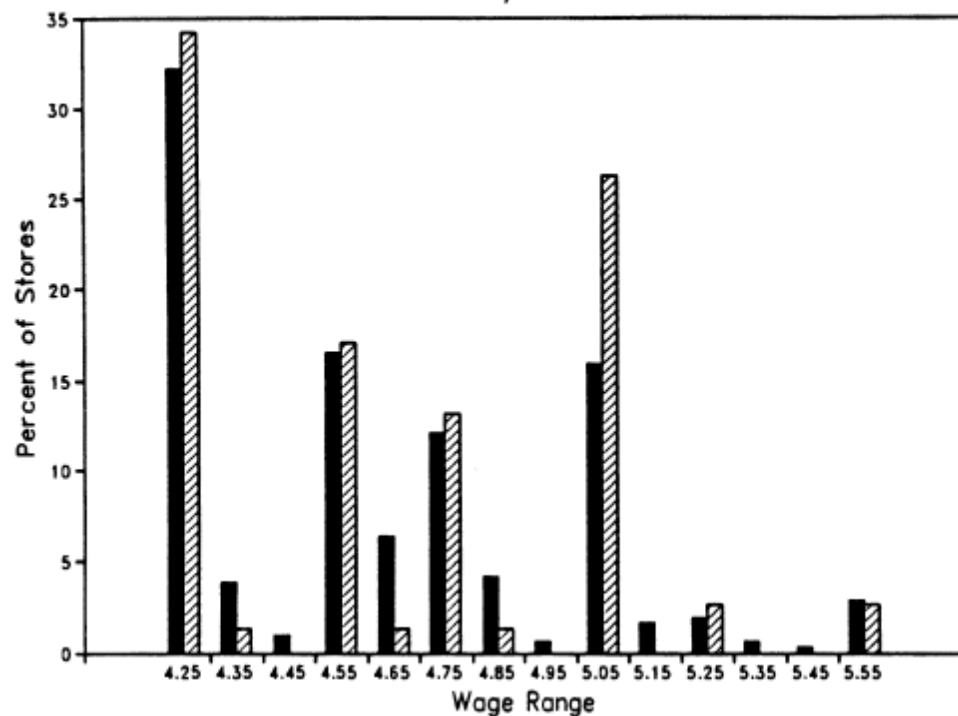
- Study the impact of increasing the minimum wage in New Jersey
- Difference-in-differences approach (check out last year’s Nobel prize!)
- **Control group:** Pennsylvania, where there was no minimum wage hike

Effect on Wages

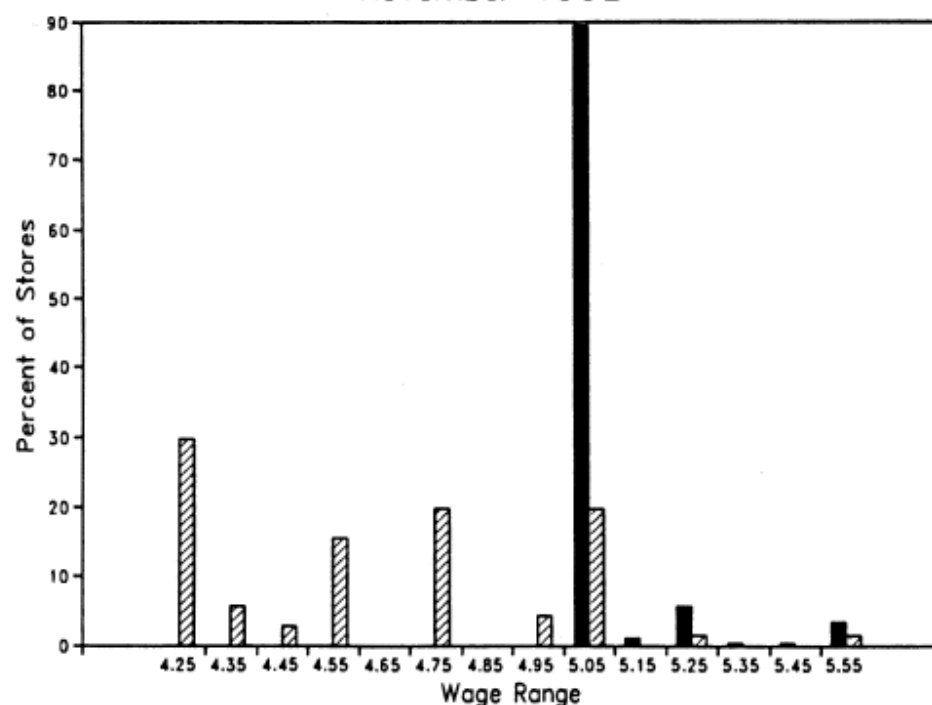


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February 1992



November 1992



■ New Jersey ▨ Pennsylvania

Consensus Breaks



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Card and Krueger (1994, 1995) and the “credibility revolution”

- Study the impact of increasing the minimum wage in New Jersey
- Difference-in-differences approach (check out last year’s Nobel prize!)
- **Control group:** Pennsylvania, where there was no minimum wage hike
- Diff-in-diff

$$\Delta E_i = a + X_i' b + c N J_i + \varepsilon_i$$

where X_i is baseline store characteristics

- Exposure design

$$\Delta E_i = \tilde{a} + X_i' \tilde{b} + \tilde{c} GAP_i + \tilde{\varepsilon}_i$$

where $GAP_i = N J_i \cdot \left\{ \frac{5.05 - W_{1i}}{W_{1i}}, 0 \right\}$

Consensus Breaks



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TABLE 4—REDUCED-FORM MODELS FOR CHANGE IN EMPLOYMENT

Independent variable	Model				
	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	2.33 (1.19)	2.30 (1.20)	—	—	—
2. Initial wage gap ^a	—	—	15.65 (6.08)	14.92 (6.21)	11.91 (7.39)
3. Controls for chain and ownership ^b	no	yes	no	yes	yes
4. Controls for region ^c	no	no	no	no	yes
5. Standard error of regression	8.79	8.78	8.76	8.76	8.75
6. Probability value for controls ^d	—	0.34	—	0.44	0.40

Notes: Standard errors are given in parentheses. The sample consists of 357 stores with available data on employment and starting wages in waves 1 and 2. The dependent variable in all models is change in FTE employment. The mean and standard deviation of the dependent variable are -0.237 and 8.825 , respectively. All models include an unrestricted constant (not reported).

^aProportional increase in starting wage necessary to raise starting wage to new minimum rate. For stores in Pennsylvania the wage gap is 0.

^bThree dummy variables for chain type and whether or not the store is company-owned are included.

^cDummy variables for two regions of New Jersey and two regions of eastern Pennsylvania are included.

^dProbability value of joint F test for exclusion of all control variables.

Consensus Breaks



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Card and Krueger (1994, 1995) and the “credibility revolution”

- Study the impact of increasing the minimum wage in New Jersey
- Difference-in-differences approach (check out last year’s Nobel prize!)
- **Control group:** Pennsylvania, where there was no minimum wage hike
- Result: the minimum wage **increases** wages and employment
- Initial reactions were not welcoming

*“Just as no physicist would claim that ‘water runs uphill’, **no self-respecting economist** would claim that increases in the minimum wage increase employment. Such a claim, if seriously advanced, **becomes equivalent to a denial that there is even minimal scientific content in economics**, and that, in consequence, economists can do nothing but write as advocates for ideological interests.”*

James Buchanan, Nobel prize in Economics

Consensus Breaks



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- Scepticism comes from the fact that the finding contradicts “basic” theories (e.g. “law of demand”)
- Dangerous implication of this argument:
 - Arguing on one hand that our simplified assumptions might be unrealistic
 - Arguing on the other hand that model predictions should be taken seriously, and that we should not trust data if our ideas are rejected

What is the Lesson?



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- Economic models are important as they can capture the most important mechanisms at play
- Simplicity is valuable
- But we should not be dogmatic about our models
- We need “credible” research designs, that could be used to reject or accept the key predictions of our models

“The idea of turning economics into a true empirical science, where core theories can be rejected, is a BIG, revolutionary idea.” - Alan Krueger (2018)

- Nobel Prize in Economics in 2021 (David Card, Joshua Angrist, Guido Imbens)



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Section 3: Empirical Designs To Assess the Impact of Minimum Wages

New Minimum Wage Research



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- New minimum wage research shifted the focus from theory to “reduced” form empirical evidence on employment and wages
- U.S. has been a fertile ground since state-level variation can be exploited
 - Two-way Fixed Effects estimation (e.g. Neumark and Wascher, 1993)

Two-way Fixed Effects estimation



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- Card and Kruger (1994) exploit one event: the New Jersey minimum wage hike in 1992
- Neumark and Wascher (1993) exploit all state-level minimum wage changes that took place between 1970 and 1989

This is a great idea:

- Most evidence before was based on time series, this is based on panel data
 - Control for state-level fixed characteristics
- Estimates based on one specific event could be misleading
- With multiple events, the biases coming from event-specific peculiarities likely to cancel each other out --> more external validity

Two-way Fixed Effects estimation



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- Change in employment of teens, by estimating the following regression (Neumark and Wascher, 1993)

$$Y_{st} = \alpha + \beta MW_{st} + X_{st}\Lambda + \gamma_s + \tau_t + \nu_{st}$$

where

- Y_{st} is the employment teen (or youth) population rate of teens at state s at time t
- MW_{st} is the level of minimum wage
- X_{st} are controls (e.g. proportion of teens in school)
- γ_s are state and τ_t time fixed effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Minimum wage	.07 (.10)	-.17 (.07)	.21 (.07)	-.11 (.05)	-.33 (.08)	-.25 (.04)	-.18 (.07)	-.23 (.04)
Proportion of population in age group	-.19 (.22)	-.11 (.15)	-.24 (.15)	-.13 (.10)	.35 (.28)	-.05 (.15)	.02 (.21)	-.16 (.11)
Prime-age male unemployment rate	-.54 (.11)	-.31 (.07)	-.86 (.08)	-.63 (.06)	-1.53 (.13)	-.76 (.07)	-1.39 (.11)	-.80 (.06)
Proportion of age group in school	...	-.75 (.03)	...	-.75 (.03)	...	-.95 (.02)	...	-.96 (.02)
Year effects	Y	Y	N	N	Y	Y	N	N
State effects	Y	Y	Y	Y	N	N	N	N
P-value for restricted model ²	...	...	.00	.00	.00	.00	.00	.00
$\bar{R}^2$	.69	.86	.68	.85	.20	.78	.18	.78
Elasticity ³	.06	-.14	.17	-.09	-.27	-.20	-.14	-.18

Should we Control for Proportion
of Teens in School?



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Should we Control for Proportion of Teens in School?



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Why control for it?

- The underlying idea is to control for changes in labour supply
- Card, Krueger, Katz (1993) criticize how the variable was measured and sensitivity of the results. Neumark and Wascher (1994) replied

Why is it a bad idea?

- A more fundamental issue is why control for changes in labour supply to begin with
- If MW affects school enrolment --> we have a “bad control” problem! (see Angrist and Pischke, “Mostly Harmless Economics”)
- If school enrolment is not reflecting the effect of MW, then it reflects “something else”, which also tends to change when states increase MW
- But that “something else” might also affect directly teen employment (omitted variable bias)

Why teens?



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The paper studies the impact on teens (16-18) and young adults (20-24)

Why teens?



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The paper studies the impact on teens (16-18) and young adults (20-24)

- Young people are low wage --> this group is heavily affected by the minimum wage

Is that an interesting group to study?

- I am not sure...
- What if teen employment falls, but prime-age adult employment increases?
- What if teen employment falls, and school attainment increases?

Still in the USA context, the teen estimates are the most common

- CBO uses the teen estimates to evaluate the potential impact of raising the minimum wages

What Relationship?



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- The paper studies the relationship between minimum wages and employment
- Is this what we are interested in?
- Neumark and Wascher (1992) calculate the

$$\text{MW elasticity} = \frac{\% \Delta \text{Employment}}{\% \Delta \text{MW}}$$

- What the theory needs instead is

$$\text{Own wage elasticity} = \frac{\% \Delta \text{Employment}}{\% \Delta \text{Wages}}$$

What Relationship?



- Usual argument in favour of MW elasticity:
- The policy maker can influence $\% \Delta MW$ and not $\% \Delta Wages$
- The policy maker should look at how changes in the policy instrument affect outcomes
- Yes, but: isn't the impact of $\% \Delta MW$ on $\% \Delta Wages$ important for the policy maker?
- The link between $\% \Delta MW$ on $\% \Delta Wages$ is also crucial for policy evaluation
 - 10% MW $\rightarrow$ 10% $\Delta Wages$ $\rightarrow$ 1% ΔEmp implies an own wage employment elasticity -0.1 (almost 0)
 - 10% MW $\rightarrow$ 1% $\Delta Wages$ $\rightarrow$ 1% ΔEmp implies an own wage employment elasticity -1 (quite large)

Own Wage Elasticity



- We are interested in how raising wages affect employment:

$$Emp_{st} = \alpha_3 + \beta_3 Wage_{st} + X_{st}\Lambda_3 + \gamma_s + \tau_t + \nu_{st}$$

- To establish that relationship, first study the relationship between minimum wages and wages:

$$Wage_{st} = \alpha_1 + \beta_1 MW_{st} + X_{st}\Lambda_1 + \gamma_s + \tau_t + \nu_{st}$$

- Then we can use the predicted wage change $Wage_{st}$
- Then we run the following regression

$$Emp_{st} = \alpha_1 + \beta_1 \widehat{Wage_{st}} + X_{st}\Lambda_1 + \gamma_s + \tau_t + \nu_{st}$$

- Does this remind you something?

Own Wage Elasticity



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- This is exactly a 2SLS estimation where MW is used as an IV

First stage: relationship between minimum wages and wages

Second stage: relationship between employment and predicted wage change

Why do we need IV to establish the relationship between wages and employment?

- **Simultaneity bias:** The fall in wages and employment might reflect a fall in labour supply
- **Omitted variable bias:** Maybe wages and employment simply reflect a general drop in labour demand (not a movement along a labour demand curve)
- **Measurement error:** Wages measured with noise

Own Wage Elasticity



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This is exactly a 2SLS estimation where MW used as an IV

- **First stage:** relationship between minimum wages and wages
- **Second stage:** relationship between employment and predicted wage change

Whether minimum wage can be used to study the employment effects is a question of whether it is a good IV or not

- Does the exclusion restriction hold?
 - MW only affects employment through changes in wages
 - This could be violated if MW coincide with other policy changes
- Is the first stage relationship strong enough?
 - If not, we face a weak IV problem

Still a hot topic issue today....

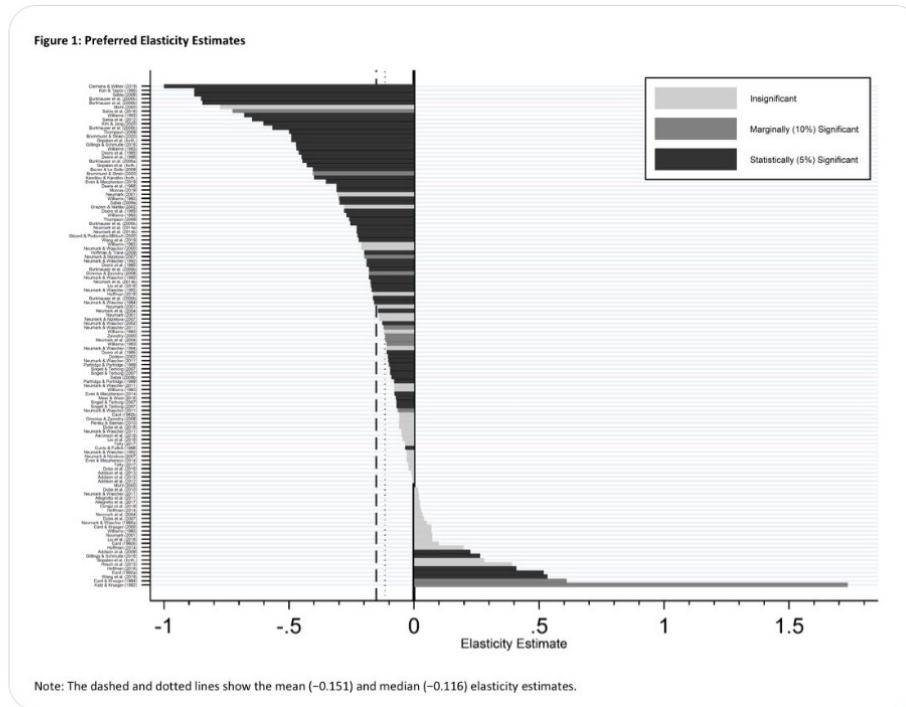


William J. Luther @WilliamJLuther · Oct 4



“The good news is we have a thick literature on the topic at this point.”

The thick literature:



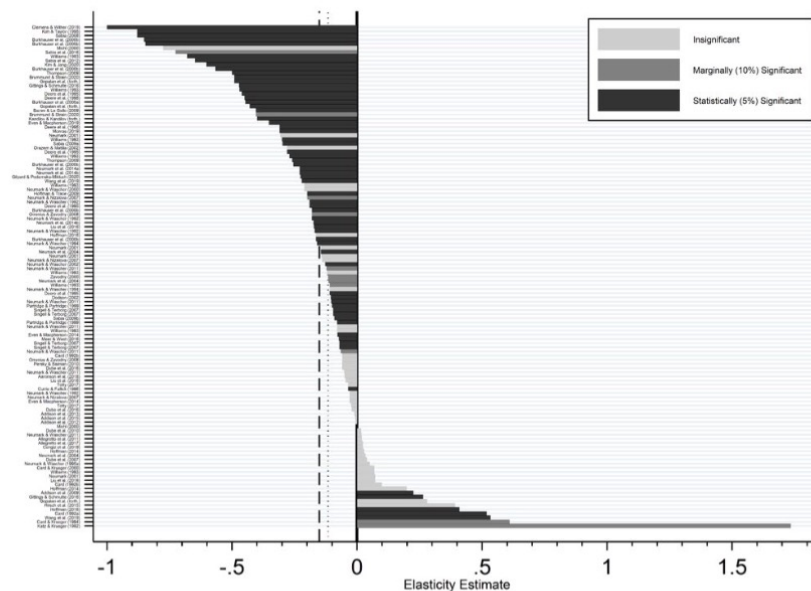
Still a hot topic issue today.



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“The good news is we have a thick literature on the topic at this point.”
The thick literature:

Figure 1: Preferred Elasticity Estimates



Note: The dashed and dotted lines show the mean (-0.151) and median (-0.116) elasticity estimates.



Arin Dube @arindube · Oct 4

If you're interested in understanding the literature studying minimum wage effects on jobs, I encourage you to look at our work on the own-wage elasticity (OWE). It is, indeed, a thick literature, and most studies suggest effects that are either positive or a small negative, where magnitudes are economically interpretable.

Own-Wage Elasticity: Quantifying the Impact of Minimum Wages on Employment

Arindrajit Dube & Ben Zipperer

NBER Working Paper 32925 · September 2024

ABSTRACT

The own-wage elasticity (OWE) of employment estimated using minimum wage increases provides an economically meaningful measure of the policy on jobs. We discuss how to interpret the magnitude of the OWE, including in terms of welfare and under alternative models of the labor market. We present a comprehensive set of OWE estimates from 88 studies and introduce a regularly updated repository of the estimates (economic.github.io/owe), an up-to-date snapshot of the existing literature for scholars and policymakers. We find that most studies to date suggest a fairly modest impact of minimum wages on jobs: the median OWE estimate of 72 studies published in academic journals is -0.13, which suggests that only around 13 percent of the potential earnings gains from minimum wage increases are offset due to associated job losses. Estimates published since 2010 tend to be closer to zero.

-0.13 **≈ 13%**

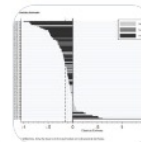
Median OWE across 72 published studies of potential earnings gains offset by job losses



Page: <https://www.nber.org/papers/w32925> · Data: Minimum Wage OWE Repository v1.0.0, economic.github.io/owe · Cts truncated at [-5, 5] (arrow)



William J. Luther @WilliamJLuther · Oct 4



“The good news is we have a thick literature on the topic at this point.”
The thick literature: x.com/arindube/status...

Fixed Effect vs. First Difference



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- One can estimate panel models using fixed effect:

$$Y_{st} = \alpha + \beta MW_{st} + X_{st}\Lambda + \gamma_s + \tau_t + \nu_{st}$$

- Or using first difference:

$$\Delta Y_{st} = \beta \Delta MW_{st} + \Delta X_{st}\Lambda + \Theta_t + \epsilon_{st}$$

First-Difference Estimates of Minimum Wage Effects
on Employment-to-Population Ratio¹

	<u>Teenagers (16-19)</u>				<u>Young Adults (16-24)</u>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Minimum wage	.23 (.12)	.00 (.09)	.38 (.10)	.10 (.07)	.15 (.08)	.01 (.07)	.27 (.07)	.07 (.06)
Proportion of population in age group	.29 (.24)	-.06 (.18)	.20 (.23)	-.10 (.18)	.20 (.11)	.00 (.09)	.09 (.10)	-.08 (.09)
Prime-age male unemployment rate	-.06 (.13)	.02 (.09)	-.46 (.10)	-.32 (.07)	-.08 (.09)	-.11 (.08)	-.45 (.07)	-.42 (.06)
Proportion of age group in school	...	-.74 (.03)	...	-.74 (.03)	...	-.71 (.04)	...	-.72 (.04)
Year effects	Y	Y	N	N	Y	Y	N	N
P-value for restricted model ²	...	...	.00	.00	...	...	.00	.00
$\bar{R}^2$	.08	.50	.05	.48	.13	.38	.07	.34
Elasticity ³	.19	.01	.31	.08	.09	.01	.17	.04

Fixed Effect vs. First Difference



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- FE and FD estimators give exactly the same coefficients if $T = 2$
- If $T > 2$, FE and FD will not be identical, but both will be unbiased, and consistent estimators
- Whenever FE and FD give you different results, hard to tell which is right

The following is recommended if T is large, N is small:

- FD usually better as FE is sensitive to small violations in assumption (we will see more on this)

The following is recommended if T is small, N is large:

- Depends on the autocorrelation of the error term. If it is small FE is better, if it is large FD is better

More Modern Approach



- Bias in the standard TWFE estimates (Callaway and Sant’Anna, 2020, de Chaisemartin and D’Haultfoeuille, 2020; Sun and Abraham, 2021; Goodman-Bacon, 2021)
 - staggered adoption
 - heterogeneous treatment effects

(Stacked) Event-study (Cengiz et al., 2019)

- Take every large minimum wage changes (event)
- Create a control group: all states without the minimum wage 3 year before and 5 year after (“event window”)

$$y_{hst} = \beta D_{hst} \times POST_h + \alpha_{hs} + \Theta_{ht} + \epsilon_{sht}$$

Event Study vs TWFE



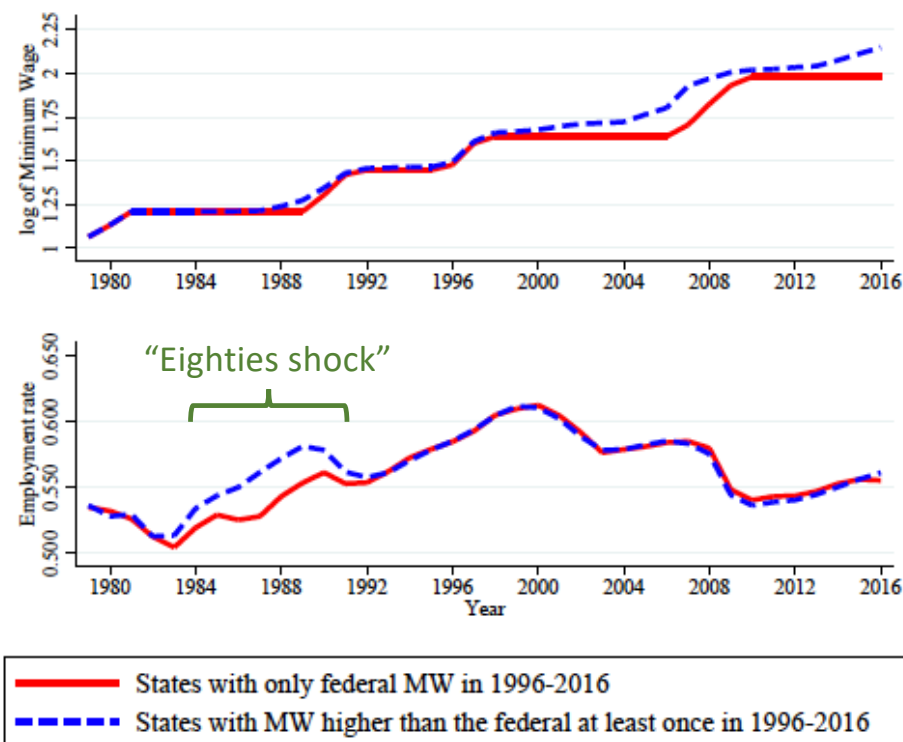
TWFE estimation leads to **biased** estimates in the US context

- “**Eighties shock**” between states with and without local minimum wages

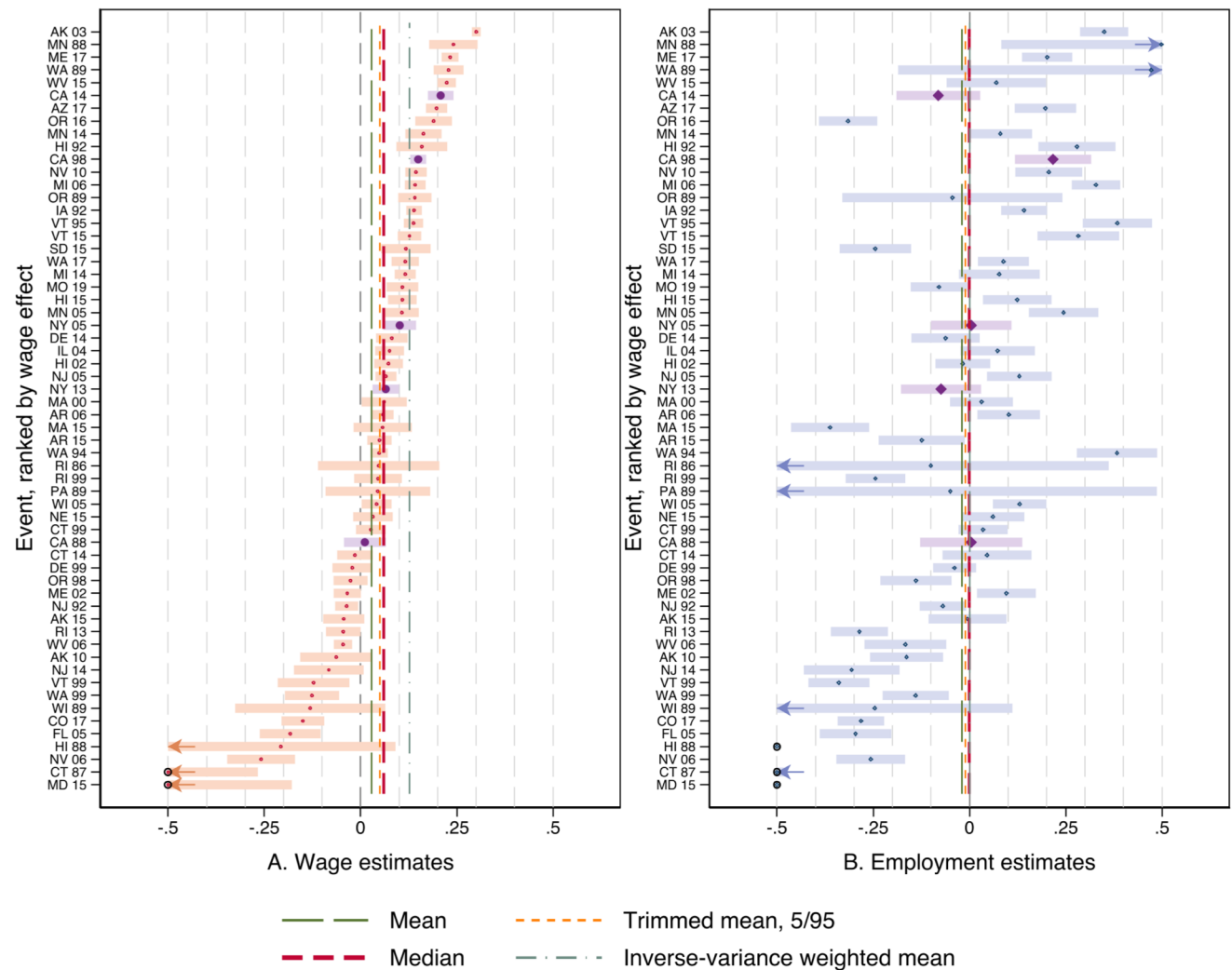
TWFE: does **all pairwise comparisons**:

- Treatment and control differences far away the events

Event study: only comparison within the event window!



Event-by-Event Estimates



New Minimum Wage Research



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- New minimum wage research shifted the focus from theory to “reduced” form empirical evidence on employment and wages
- U.S. has been a fertile ground since state-level variation can be exploited
 - Two-way Fixed Effects estimation (e.g. Neumark and Wascher, 1993)
 - Usage of administrative data (e.g. Card and Krueger, 2000)
 - Border-discontinuity design (e.g. Dube, Lester and Reich, 2010)

Leveraging proximity and the border discontinuity design



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- TWFE approach compares the employment change across states with vastly different economic and political environments
- Time and state fixed effects absorb the time-invariant differences, but still the differences can be large especially if we study long period
- A key idea of Card and Krueger (1994) is the leverage of the proximity between PA and NJ:
- These states are very similar politically and economically, but one of them has a MW increase
- Dube, Lester, Reich (2010) extend this proximity idea by exploiting many MW changes over time.

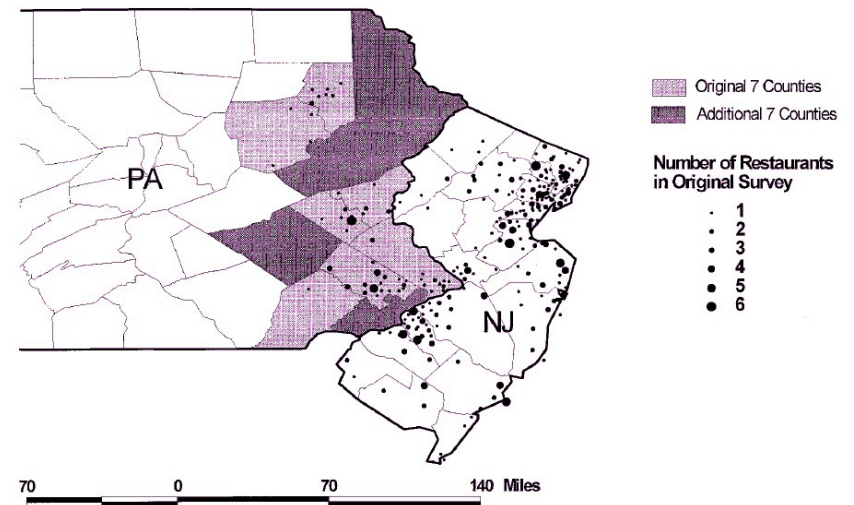
Leveraging proximity: Card and Krueger (1994, 2000)

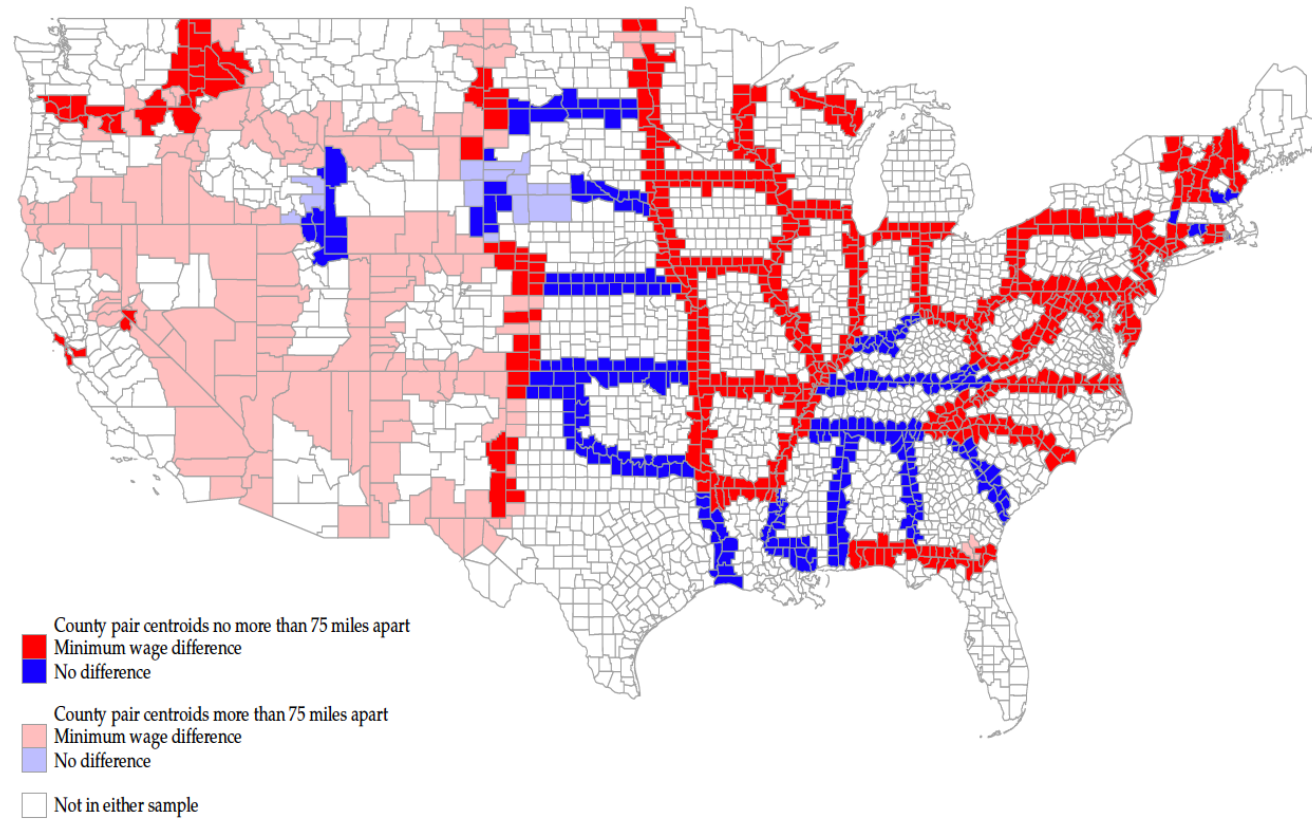


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Card and Krueger (1994, American Economic Review) study

- NJ and PA fast food restaurants NJ raised minimum wage, PA did not
- Self-collected survey
- Small positive/no effect on jobs
- Revisit (2000, AER) using representative payroll records from UI filings → No effect on jobs





US border county sample (2000-2011)

Leveraging proximity: Card and Krueger (1994, 2000)



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- Canonical two-way FE estimator:

$$Y_{jt} = \alpha + \beta MW_{jt} + X_{jt}\Lambda + \gamma_j + \tau_t + \nu_{jt}$$

- County-pair database: stack by pairs → A county can be part of multiple pairs
- Pair-specific fixed effects

$$Y_{jt} = \alpha + \beta MW_{jt} + X_{jt}\Lambda + \gamma_j + \tau_{pt} + \nu_{jpt}$$

- Wash out variation between pairs; only use within-pair variation

QWI estimates - Teen (2000-2011)

	Teens	
	(1)	(2)
<i>Earnings</i>	0.177*** (0.036) 83,462	0.222*** (0.047) 83,462
<i>Employment</i>	-0.173** (0.071) 84,702	-0.059 (0.084) 84,702
<i>Hires</i>	-0.515*** (0.094) 80,944	-0.219** (0.094) 80,944
<i>Separations</i>	-0.552*** (0.100) 74,952	-0.233** (0.098) 74,952
<i>Turnover Rate</i>	-0.377*** (0.061) 74,509	-0.204*** (0.072) 74,509
<u>Controls:</u>		
Common time effects	Y	
Pair-specific time effects		Y

Checking parallel trends assumption with distributed lags



$$Y_{jt} = \sum_{k=-A}^B \beta_k MW_{i,t-k} + X_{it}\Lambda + \gamma_i + \tau_{gt} + \nu_{it} \quad (3)$$

- β_k are responses to specific leads and lags
 - A leads, B lags
 - Note that β_{-1} is the coefficient associated with a 1-unit leading $MW_{j,t+1}$
 - Cumulative response: $\eta_\tau = \sum_{k=-A}^\tau \beta_k$ traces out employment time path prior to a unit MW shock
- **Falsification test:** check values of η_k (or β_k) for $k < 0$ (prior to treatment).
- Estimates employment differential in treated v.s. control units prior to treatment
 - Common practice with DID

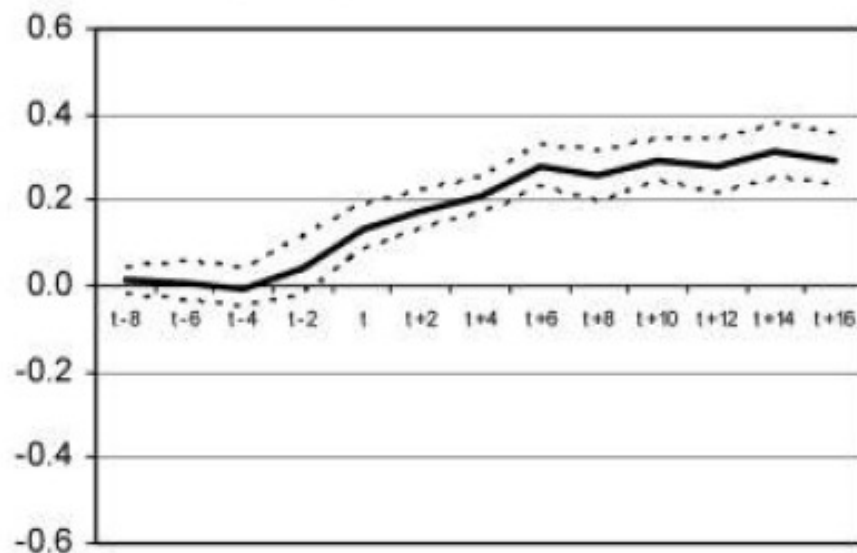
Pre-existing trends: Standard TWFE



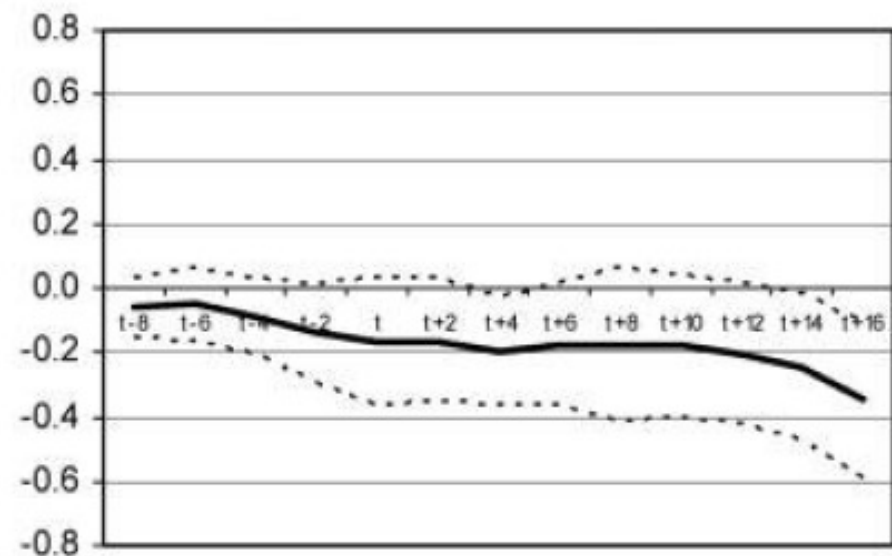
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Ln Earnings

1. All County Sample, Common Period Effects



Ln Employment

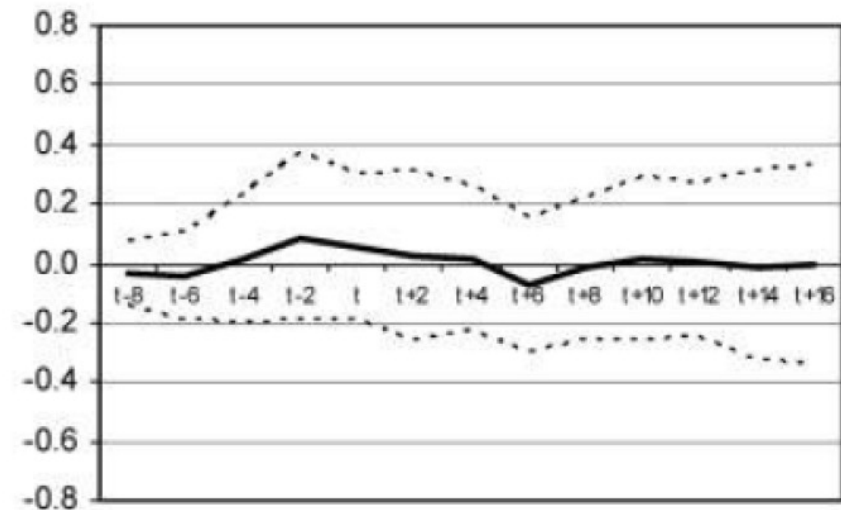
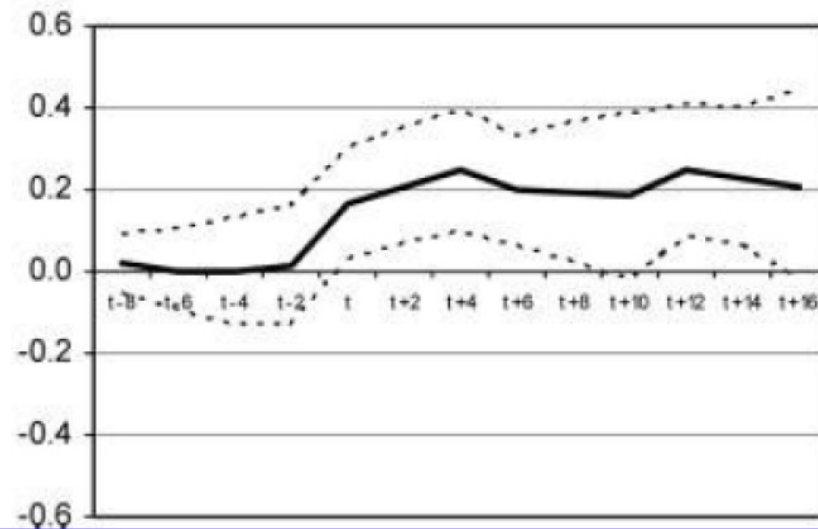


Pre-existing trends: Border Discontinuity



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6. Contiguous Border County-Pair Sample, County-Pair Specific Period Effects



Validity of the border discontinuity design



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- What's the unit of treatment? More convincing when it's not local minimum wage set at the state level
- If policy is endogenous, it's less likely when comparing adjacent counties
- City versus state-level variation

Cross-border spillovers

- Outcomes may be affected on other side of the border → would bias the estimates
- We can check this: compare interior counties to border counties on both sides of the border
- Use “donut holes” similar to RDD (leave out border “rung”, but take the next closest “rung”)

New Minimum Wage Research



UCL

- New minimum wage research shifted the focus from theory to “reduced” form empirical evidence on employment and wages
- U.S. has been a fertile ground since state-level variation can be exploited
 - Two-way Fixed Effects estimation (e.g. Neumark and Wascher, 1993)
 - Usage of administrative data (e.g. Card and Krueger, 2000)
 - Border-discontinuity design (e.g. Dube, Lester and Reich, 2010)
- But most of the evidence focuses on specific demographic groups (e.g. teens) or sectors (e.g. restaurants)
 - It is possible that MW increases employment in some sectors and decreases it in others

Overall employment or teens



- Most studies in the existing US literature focus on teens or restaurant workers
- Teen share is falling over time
- Restaurant sector might be special
- What about *overall* employment?
 - Example: Meer and West (2016)
 - But for state-level aggregate workforce, there is no first-stage wage effects in recent decades

A number of recent papers to overcome this challenge

Table 1: Composition of MW workers in US over time

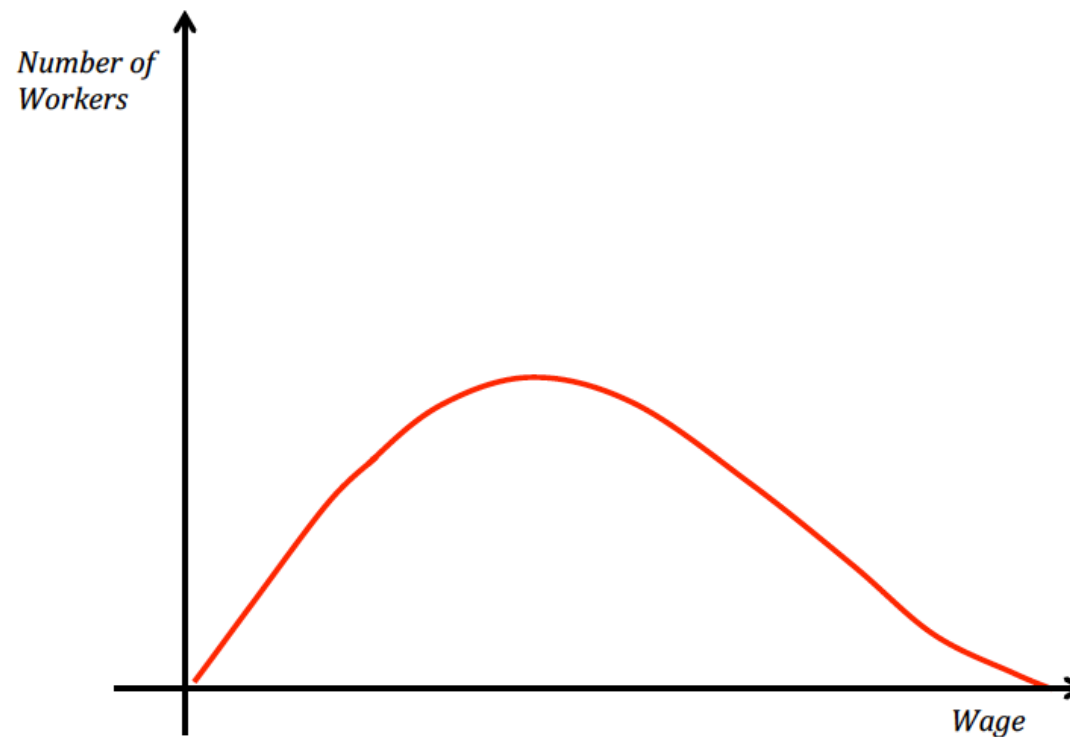
	1979-1980	1989-1990	1999-2000	2009-2010	2018-2019
<u>Industry composition</u>					
agriculture	0.04	0.04	0.04	0.03	0.04
manufacturing	0.14	0.09	0.08	0.05	0.04
retail and hospitality	0.47	0.59	0.57	0.58	0.56
health and education	0.17	0.12	0.15	0.15	0.16
other	0.18	0.15	0.16	0.18	0.20
<u>Age composition</u>					
16-19	0.28	0.29	0.27	0.18	0.18
20-24	0.22	0.21	0.20	0.24	0.22
25-30	0.13	0.14	0.13	0.15	0.15
31-40	0.14	0.16	0.18	0.16	0.16
41-50	0.11	0.10	0.13	0.14	0.13
51-64	0.13	0.09	0.10	0.13	0.16
<u>Urban composition</u>					
urban	0.65	0.71	0.75	0.81	0.87
<u>Location composition</u>					
northeast	0.20	0.12	0.14	0.14	0.17
midwest	0.25	0.26	0.19	0.24	0.20
south	0.39	0.38	0.38	0.38	0.28
west	0.15	0.24	0.29	0.24	0.35
<u>Skill composition</u>					
less than high school	0.40	0.40	0.39	0.26	0.23
high school	0.40	0.38	0.32	0.34	0.33
some college	0.15	0.17	0.24	0.31	0.31
college	0.05	0.05	0.06	0.09	0.13
<u>Race/ethnicity composition</u>					
black	0.13	0.13	0.12	0.11	0.11
hispanic	0.07	0.15	0.23	0.25	0.29
white	0.84	0.83	0.82	0.81	0.76
<u>Share MW within all workers</u>					
share	0.27	0.14	0.14	0.19	0.15

Frequency Distribution-Based Approach



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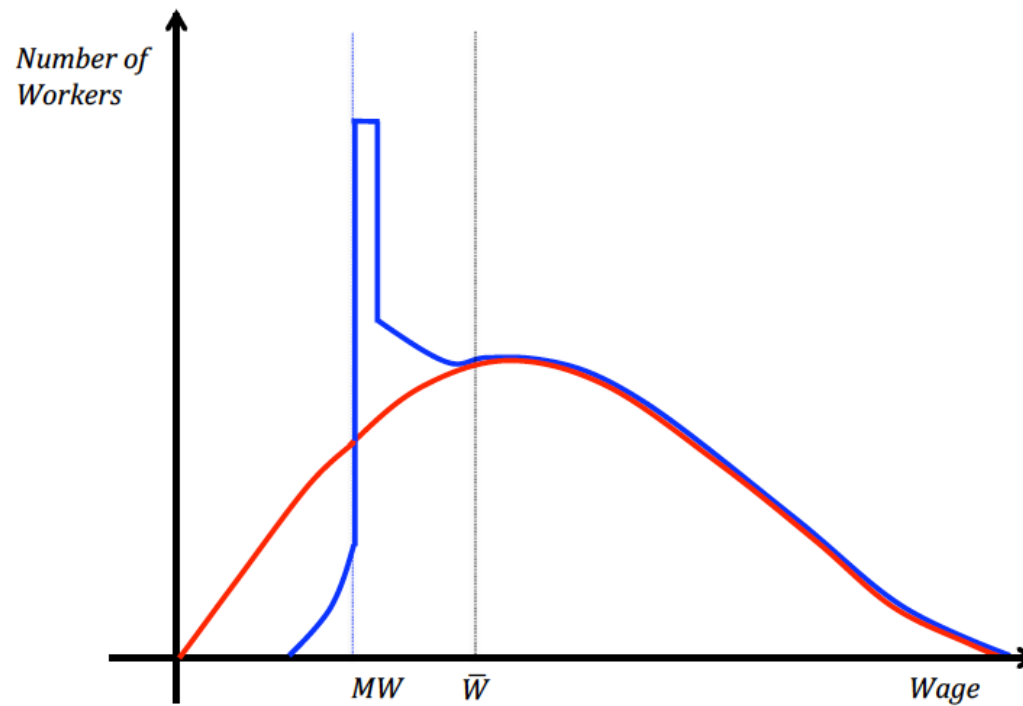
Cengiz et al. (2019) develop a novel method to assess the overall employment change



Frequency Distribution-Based Approach



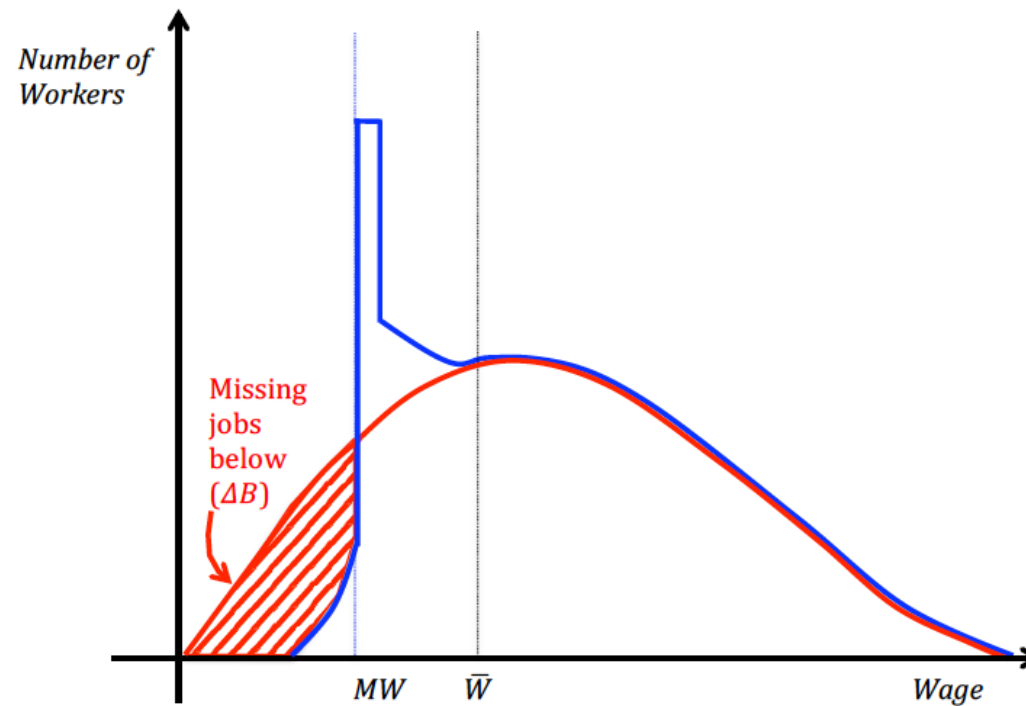
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Frequency Distribution-Based Approach



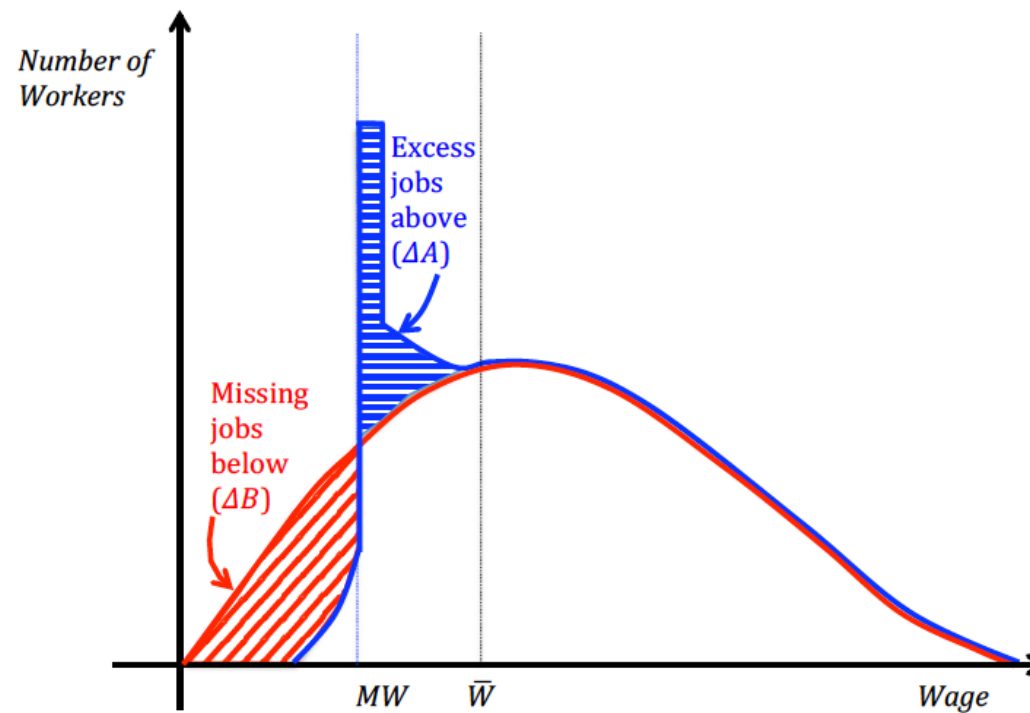
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Frequency Distribution-Based Approach



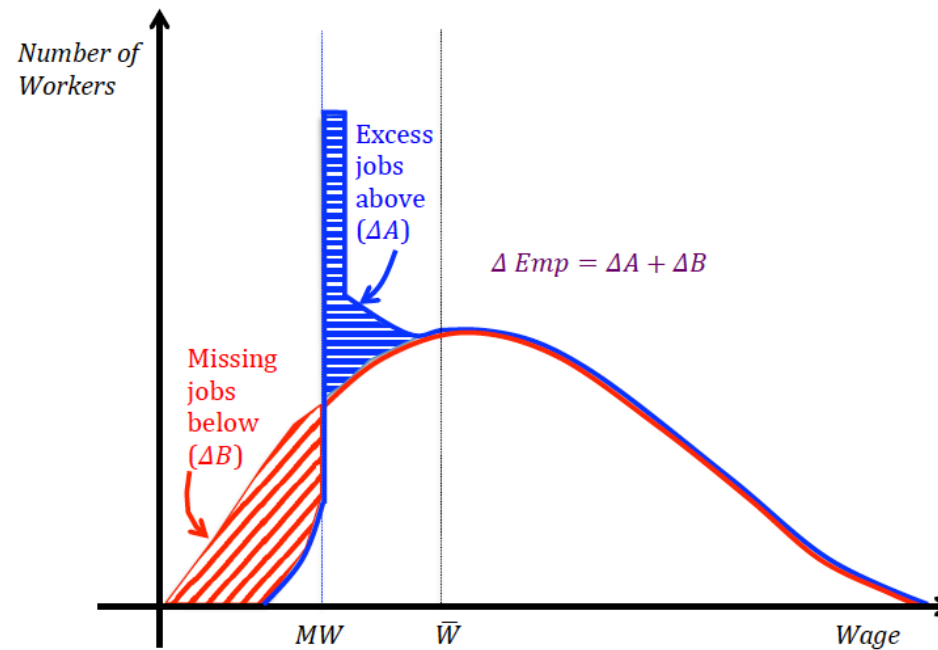
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Frequency Distribution-Based Approach



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Assess the change in missing and excess jobs

Frequency Distribution-Based Approach



UCL

- Key empirical challenge: how to get the counterfactual wage distribution
- Apply a Diff-in-Diff idea
- Case study: Washington State
- During 1999-2000, WA raised its MW from \$7.51 to \$9.18 (in 2016)
- A large, 22% increase in MW
- WA has administrative data on hourly wages

Frequency Distribution-Based Approach



UCL

- Key empirical challenge: how to get the the counterfactual wage distribution
- Apply a Diff-in-Diff idea
- Case study: Washington State
- During 1999-2000, WA raised its MW from \$7.51 to \$9.18 (in 2016)
- WA has administrative data on hourly wages

Actual number of jobs in wage bin k (post MW increase):

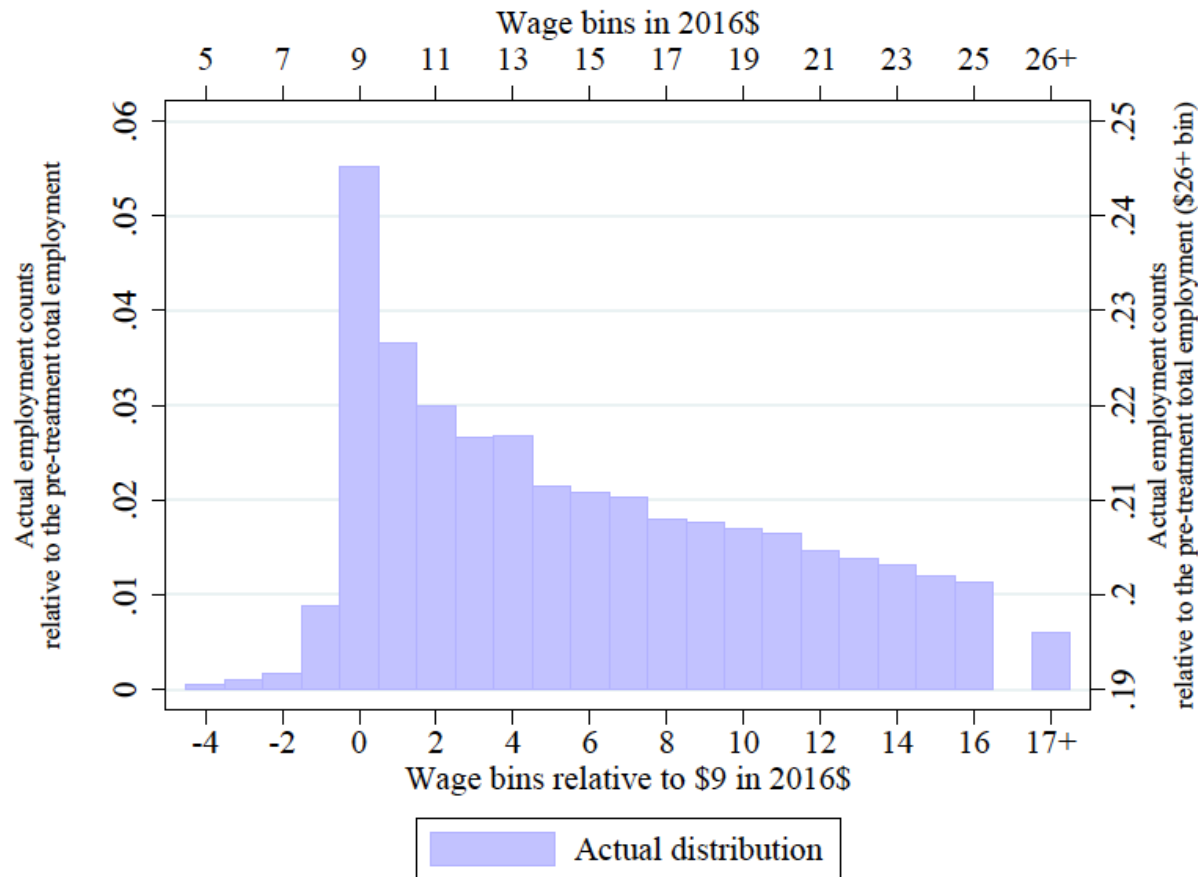
$$\Theta \times e_{WA,k,Post} = \frac{1}{\frac{E_{WA,Pre}}{N_{WA,Pre}}} \times \frac{E_{WA,k,Post}}{N_{WA,Post}}$$

where $\Theta = \frac{N_{WA,Pre}}{E_{WA,Pre}}$ is used to express job counts in pre-treatment WA aggregate employment

Frequency Distribution-Based Approach



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Frequency Distribution-Based Approach



Counterfactual number of jobs in wage bin k (post MW increase):

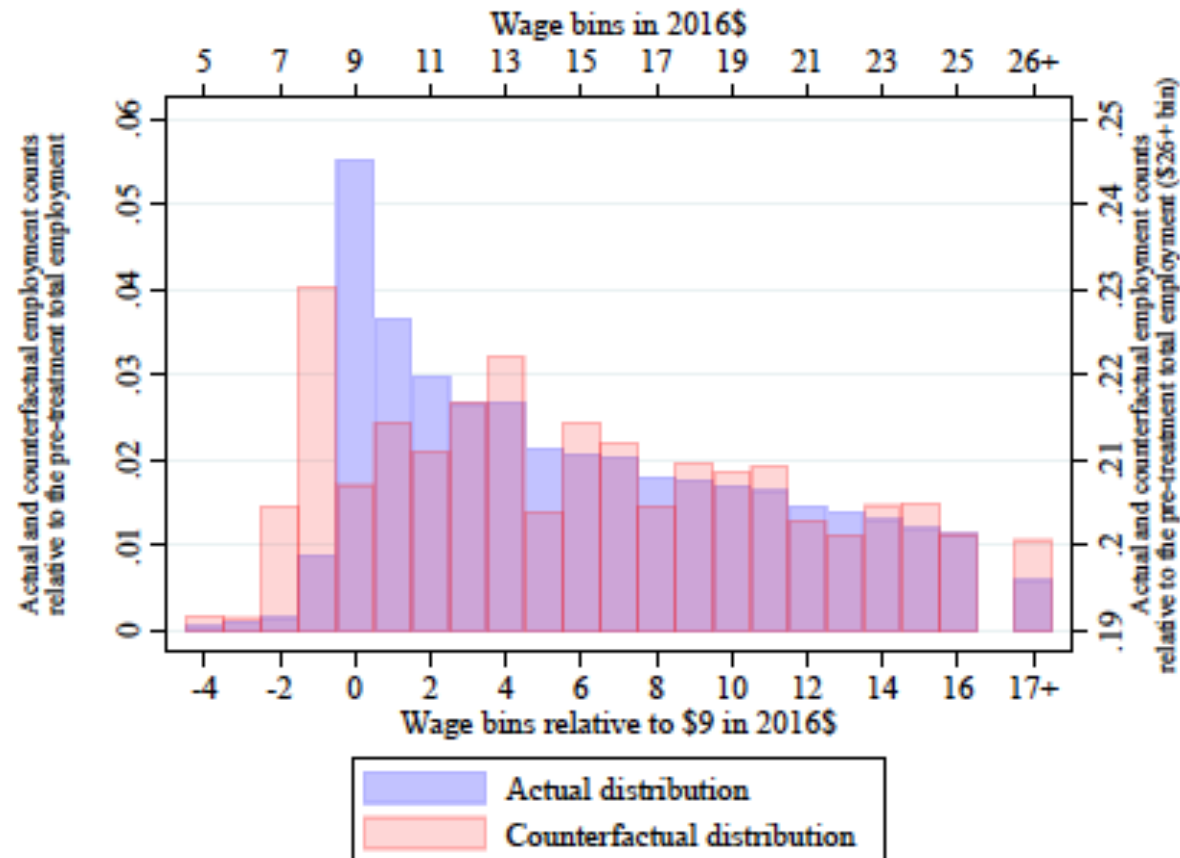
$$e_{WA,k,Post}^{CF} = \Theta \left[\underbrace{e_{WA,k,Pre}}_{\text{Pre-treatment in WA}} + \underbrace{\sum_{s \in \text{Control}} \frac{1}{39} (e_{s,k,Post} - e_{s,k,Pre})}_{\text{Change in control states}} \right]$$

where $\Theta = \frac{N_{WA,Pre}}{E_{WA,Pre}}$ is used to express job counts relative to the pre-treatment aggregate employment in WA

Frequency Distribution-Based Approach



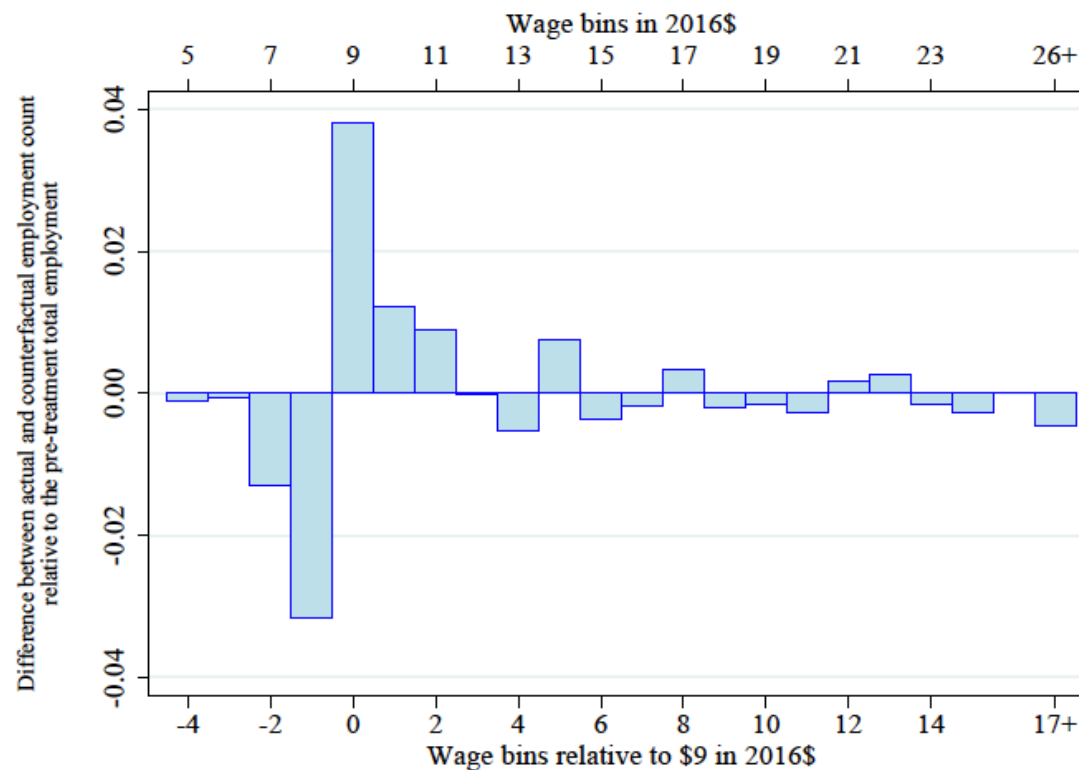
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Frequency Distribution-Based Approach



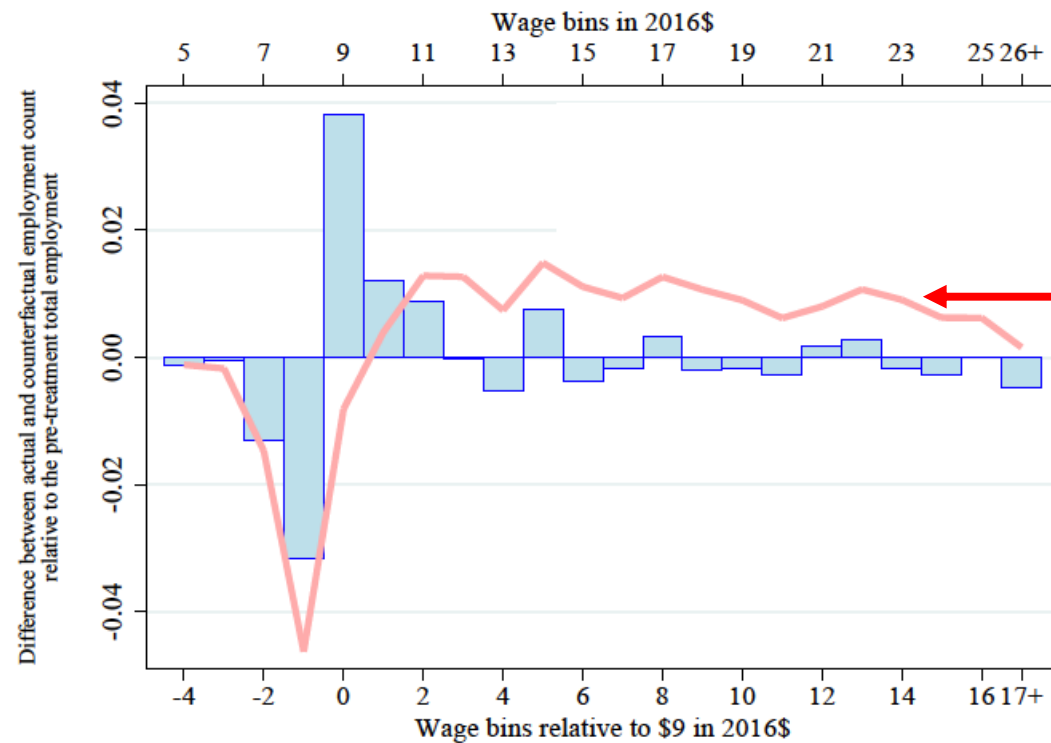
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Frequency Distribution-Based Approach



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Cumulative
employment
changes up to that
wage bin

Frequency Distribution-Based Approach



- Apply the same idea in a difference-in-differences style estimation
- Exploit 136 state-level minimum wage changes
- Use wage bin-by-state-by-period state-panel data from 1979-2016

$$\frac{E_{swt}}{N_{st}} = \sum_{\tau=-3}^4 \sum_{k=-4}^{17} \alpha_{\tau k} \text{*Treat*}_{swt}^{\tau k} + \mu_{sw} + \rho_{wt} + u_{swt}$$

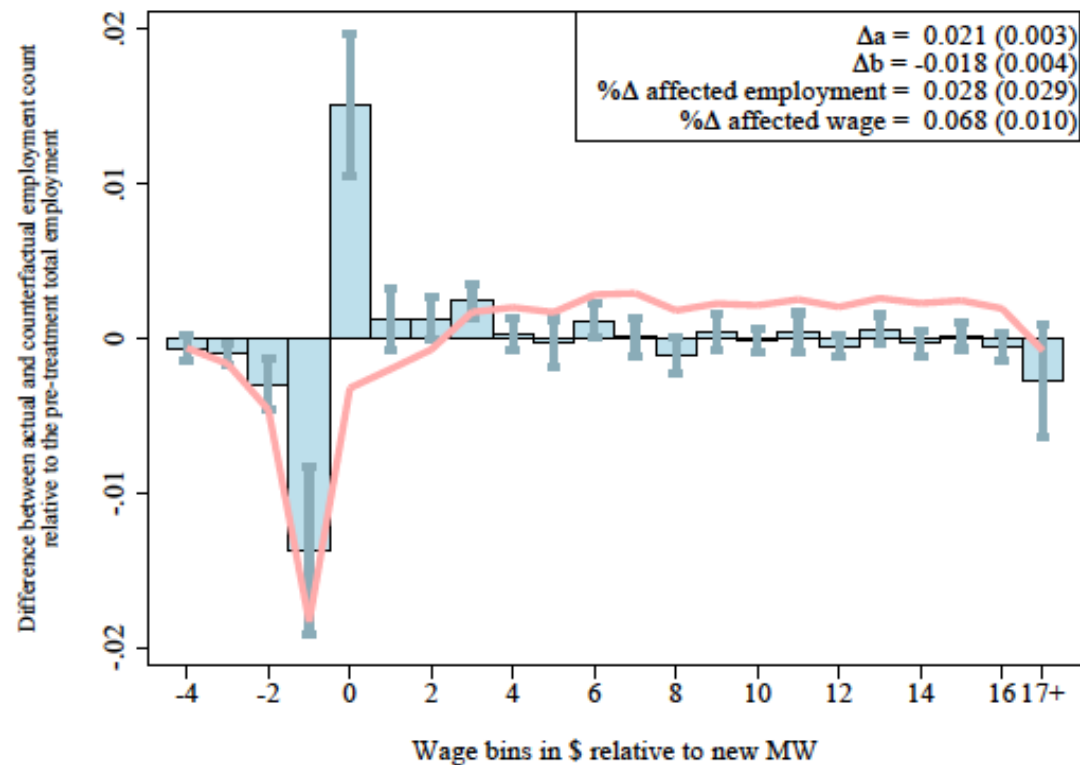
*Treat*_{swt}^{τk} = 1 for the τth year following an event, in wage bins *k* dollars from the new *MW*; 0 otherwise
e.g., *Treat*_{swt}¹¹ = 1 for first 1 year after event, in wage bins within \$1 at/above the new *MW*

Frequency Distribution-Based Approach



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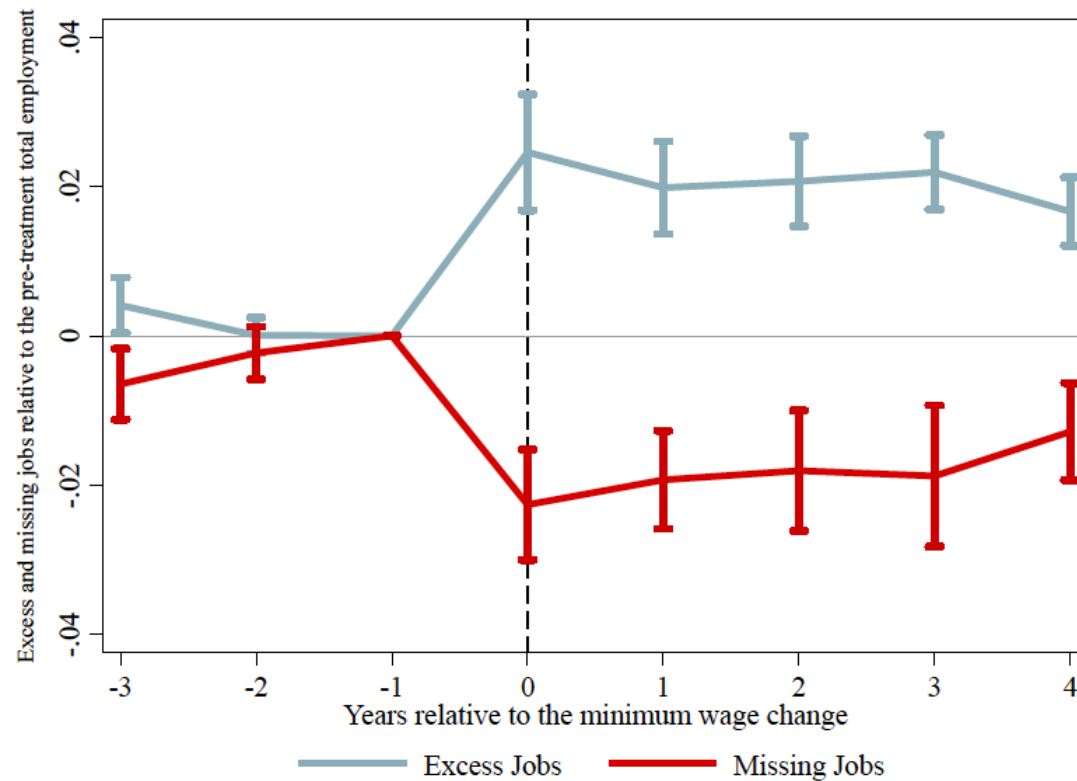
Figure 2: Impact of Minimum Wages on the the Wage Distribution



Frequency Distribution-Based Approach



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Summary: Frequency Distribution-Based Approach



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Three key advantages:

1. The localized approach is crucial for uncovering meaningful “first stage” wage effects of the minimum wage if someone studies overall employment
2. Demonstrate the changes in the upper tail - can be used for falsification test of the empirical design
3. Improve precision by cutting off the noise coming from the upper tail

New Minimum Wage Research



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- New minimum wage research shifted the focus from theory to “reduced” form empirical evidence on employment and wages
- U.S. has been a fertile ground since state-level variation can be exploited
 - Two-way Fixed Effects estimation (e.g. Neumark and Wascher, 1993)
 - Usage of administrative data (e.g. Card and Krueger, 2000)
 - Border-discontinuity design (e.g. Dube, Lester and Reich, 2010)
 - Frequency Distribution-Based Approach (e.g. Cengiz, Dube, Lindner, Zipperer, 2019)
 - Machine learning based approach (e.g. Cengiz, Dube, Lindner, Zentler-Munro, 2022)

Machine Learning-Based Approach



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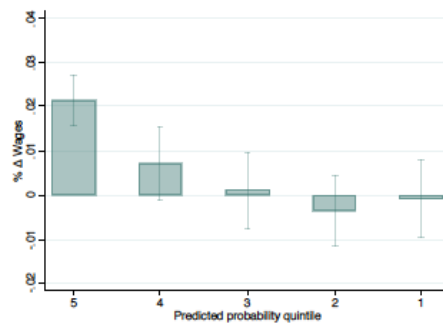
- Who is a worker affected by the minimum wage?
 - Ideally, we want to identify workers **who would have been employed at or near the minimum wage in the absence of the policy**
- A standard approach:
 - Use workers who had a minimum wage job
 - Misses the new entrants
- Use pre-determined demographic characteristics (X) to predict who is likely to be a minimum-wage worker:
$$\Pr(\text{Minimum-wage worker} \mid X) = f(X)$$
- Apply modern machine learning tools to find the best $f()$ function
- Study the impact of the minimum wage separately for each predicted probability group

Machine Learning-Based Approach

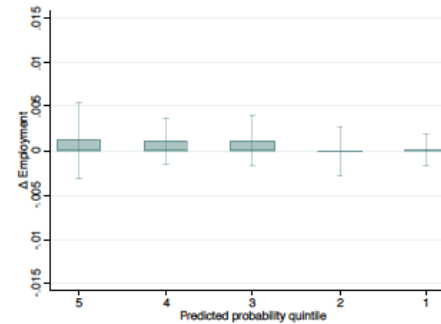


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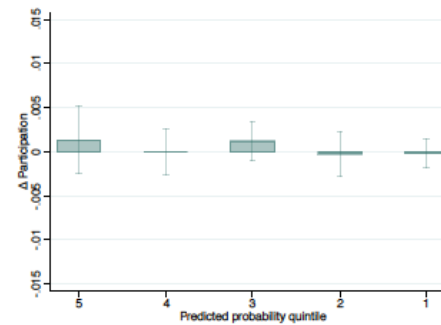
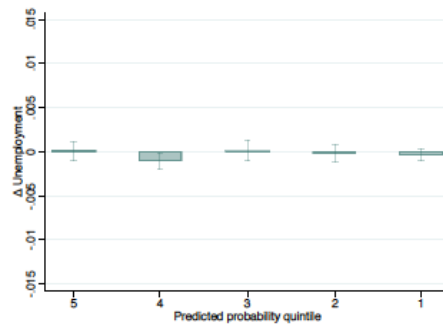
Figure 8: Impact of the Minimum Wage by Predicted Probability Quintiles



(a) Wage



(b) Employment





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Section 5: Monopsonistic Competition

Monopsonistic Competition



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Strong indication that employment effects are limited?

But why?

A model of monopsonistic competition:

- Workplaces are differentiated (and not homogenous), like products on the market
- The differentiation leads to market power, like “monopolistic competition” in the product market

The idea:

- Since some workers want to work at my firm, I can hire them even if I pay lower wages
- This creates a wedge between productivity and wages
- This means I can “exploit” workers by paying them a wage lower than the value of their work
- In equilibrium, I hire fewer workers and pay them less

Firm's Problem in the Presence of Labor Market Power



- Firms maximize their profits:

$$\max_{L_j} pf(L_j) - w(L_j)L_j$$

- The FOC:

$$pf'(L_j) - w(L_j) - w'(L_j)L_j = 0$$

or

$$pf'(L_j) = w(L_j) + w'(L_j)L_j$$

or

$$pf'(L_j) = w(L_j) \left(1 + \frac{w'(L_j)L_j}{w(L_j)} \right)$$

Firm's Problem in the Presence of Labor Market Power



- The Elasticity of Firm-level Labor Supply is defined as:

$$\eta = \frac{\partial L_j}{\partial w} \frac{w}{L_j} = \frac{1}{\frac{\partial w}{\partial L_j} \frac{L_j}{w}} = \frac{1}{\frac{w'(L_j)L_j}{w(L_j)}}$$

Note that: $\eta > 0$

Firm's Problem in the Presence of Labor Market Power



- In the presence of imperfect competition, we then have:

$$\underbrace{pf'(L_j)}_{\text{Marginal Revenue Product}} = \underbrace{w(L_j) \left(1 + \frac{1}{\eta}\right)}_{\text{Marginal Cost}}$$

- Compare this to the neoclassical framework FOC!

$$pf'(L_j) = w$$

Firm's Problem in the Presence of Labor Market Power



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- Mark-down:

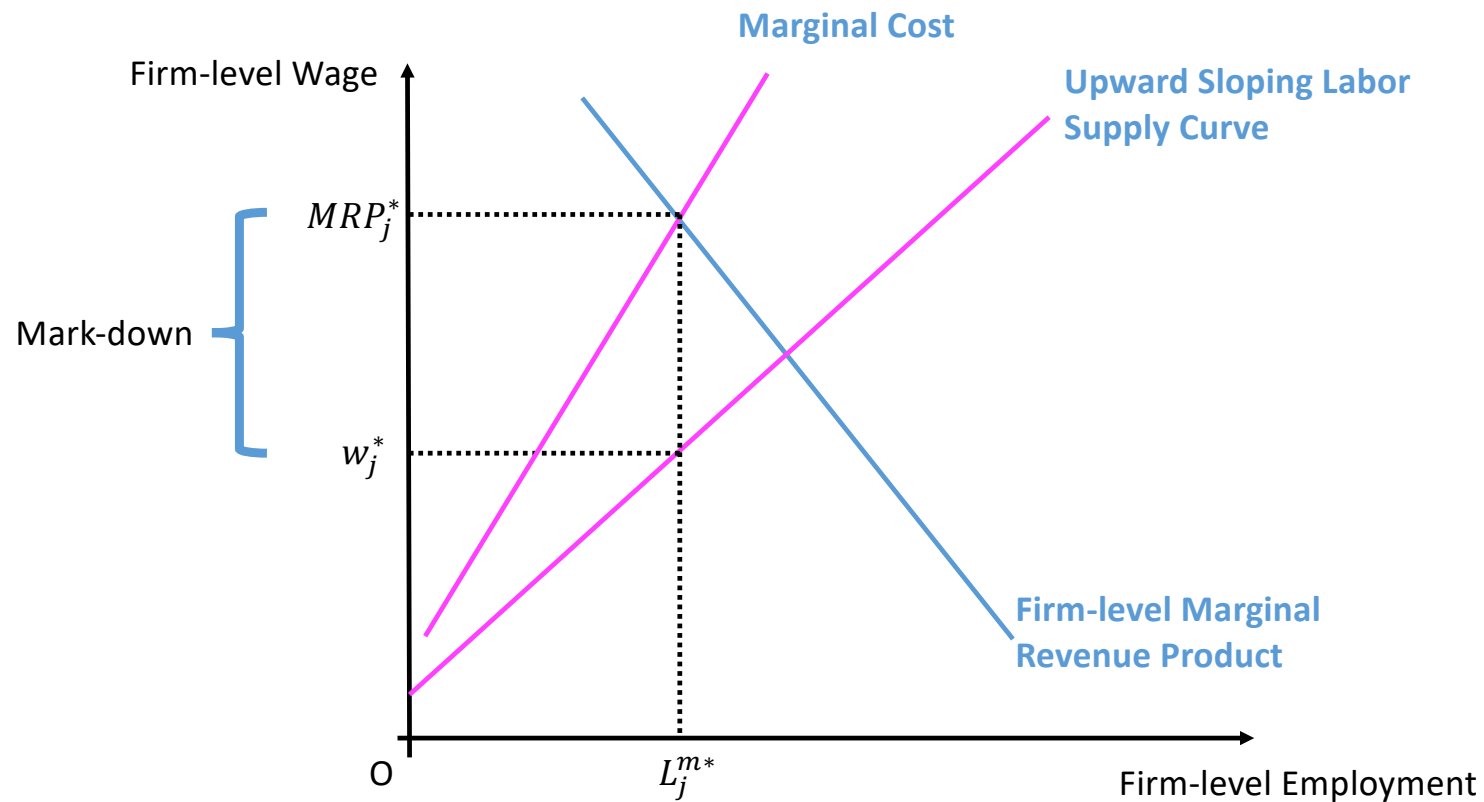
$$\underbrace{pf'(L_j)}_{\substack{\text{Marginal Revenue} \\ \text{Product}}} \cdot \underbrace{\left(\frac{1}{1 + \frac{1}{\eta}}\right)}_{\text{Mark-down}} = \underbrace{w(L_j)}_{\text{Wages}}$$

- In the presence of labour market power, there is a wedge between **marginal revenue product** and **wages**
- Firms earn “extra profit” on each worker
- The extra profit depends on η – the firm-level elasticity of labour supply
- If η is large $\rightarrow$ we get back the neoclassical framework

Each Employer Operates as a Small Monopsony



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The Model of Workplace Differentiation



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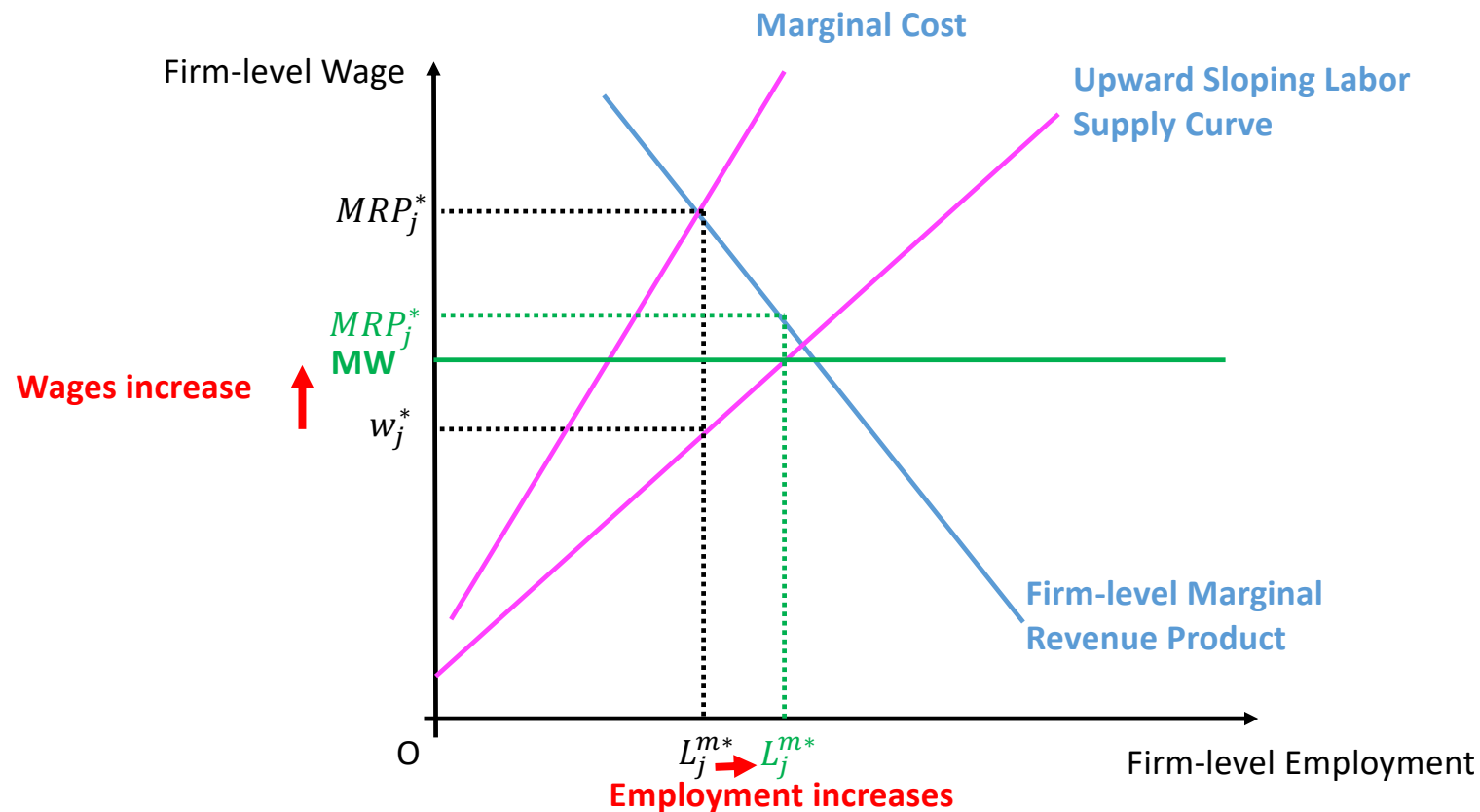
Main predictions of the model:

- People feel they are underpaid at their workplace relative to their true productivity – wages are “marked-down”
- People feel they could find a better-paying workplace, but those are not ideal (they are too far from home, not nice co-workers)
- Workers with similar skills and abilities are paid differently at different firms (there is a firm-level skill premium)
- Efficient firms are larger and pay more to workers with similar skills

Each Employer Operates as a Small Monopsony



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Section 6: Margins of Adjustment

Employment



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No significant reduction on employment + Large increase in workers' compensation



Big increase in Total Low Wage Income

Natural question: **who pays for the minimum wage?**

Who Pays For the Minimum Wage?



UCL

1. Wage retrenchment of higher-skilled workers
2. Passthrough to firm owners
3. Passthrough to suppliers
4. Passthrough to consumers
5. Improving efficiency (reorganization or better allocation)

Who Pays For the Minimum Wage?



UCL

1. Wage retrenchment of higher-skilled workers **No**
2. Passthrough to firm owners **No/Modest**
3. Passthrough to suppliers **Probably no, but little evidence**
4. Passthrough to consumers
5. Improving efficiency (reorganization or better allocation)

Who Pays For the Minimum Wage?



UCL

1. Wage retrenchment of higher-skilled workers **No**
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3. Passthrough to suppliers **Probably no, but little evidence**
4. **Passthrough to consumers Yes**
5. Improving efficiency (reorganization or better allocation)

Effect on Prices



Table 3: Evidence on Prices

Study	Industry	Effect on Prices	Pass-through
Katz and Krueger (1992)	Fast-food industry	No effect but imprecisely estimated	-
Card and Krueger (1995)	Fast-food restaurants in Texas	mixed evidence, imprecise estimates	-
Aaronson (2001)	Restaurant	Increase	Pass-through depends on specification
Dube et al. (2007)	Restaurant	Increase for fast-food restaurants	(More than) full pass-through
Aaronson et al. (2008)	Restaurant	Increase for fast-food restaurants	(More than) full pass-through
Basker and Khan (2016)	Fast-food industry	Increase	Sizeable pass-through
Allegretto and Reich (2018)	Restaurant	Increase	Nearly full pass-through
Harasztosi and Lindner (2019)	Manufacturing (for prices)	Increase	Partial pass-through (75%)
Ashenfelter and Jurajda (2022)	Fast-food industry	Increase	Full pass-through
Leung (2021)	Retail	Increase in grocery stores	(More than) full pass-through
Renkin et al. (2022)	Retail	Increase of grocery prices	Full pass-through

Many studies on price adjustment:

- Evidence almost exclusively suggests that minimum wages **passed through to consumers**
- Worth keeping in mind that the price changes are not quality adjusted!

Review of the Evidence



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Profit changes

- Stock-market value (reflecting financial market reactions)
 - Card and Krueger (1995) find no effect, Bell and Machin (2018) large effect
- Profitability (Profit/Revenue)
 - Draca, et al. 2011 in UK find a reduction, while Mayneris et al., (2016) find no change
 - Note that the fall in revenue could reflect a fall in profit or increase in revenue
- Profit (relative to past revenue or profit per worker)
 - Harasztosi and Lindner (2019) in Hungary; Vergara, WP 2023 in the US find a modest fall in profit (around 20% of the increase in labour cost)
 - Drucker et al. (2021) find a significant reduction in profits in Israel

Case Study: Hungary



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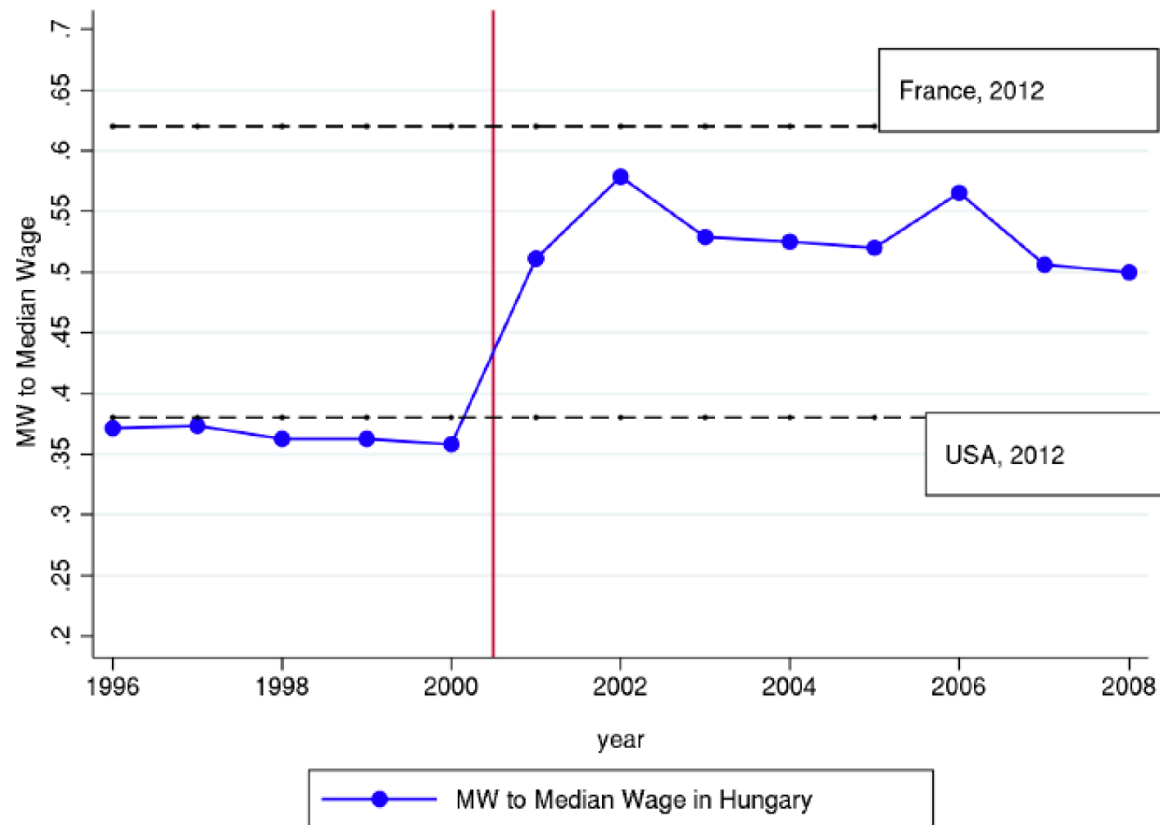
- US min wage variation tends to be small and short-run in nature
- Sorkin (2018): small employment elasticities in the USA context consistent with large long-term employment changes
- To deal with this issue: look at the effect of large and persistent changes in the minimum wages

A case study from Hungary!

Now we are talking...



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Incidence of Minimum Wages

- ▶ Our starting point is the following accounting identity:

$$Profit \equiv Revenue - Material - LaborCost - Depr - MiscItems$$

- ▶ Estimate the following equation

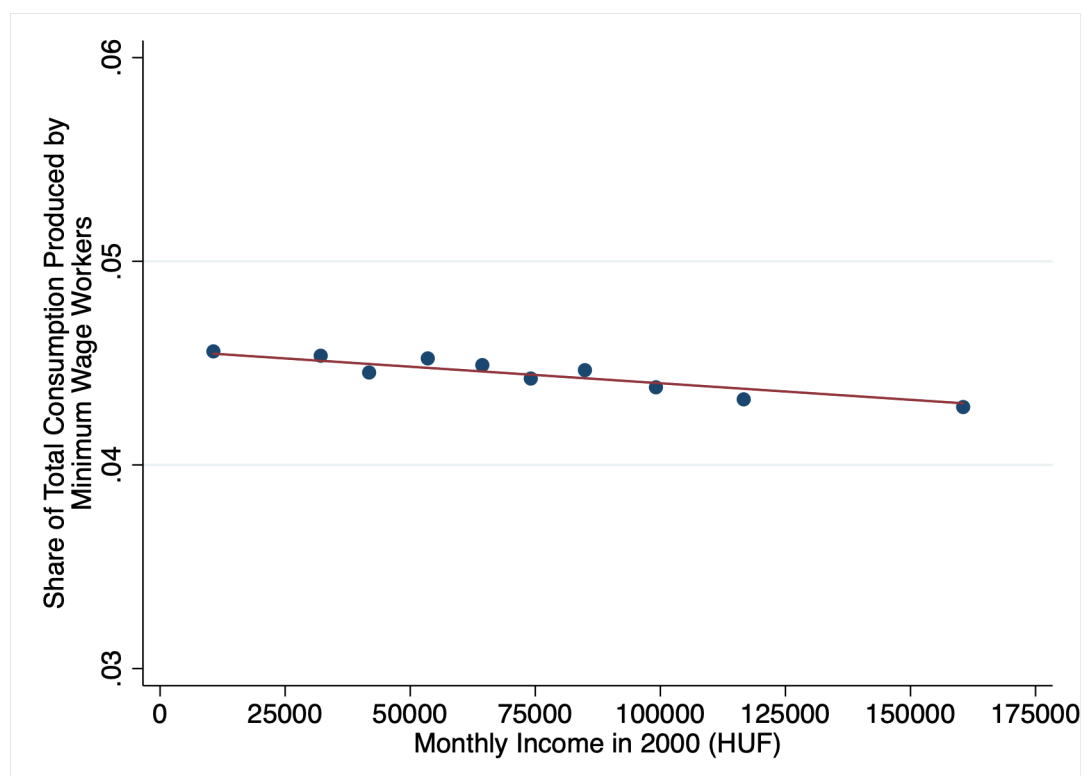
$$\frac{\Delta LaborCost}{Revenue_{2000}} = \underbrace{\frac{\Delta Revenue}{Revenue_{2000}} - \frac{\Delta Material}{Revenue_{2000}} - \frac{\Delta MiscItems}{Revenue_{2000}}}_{\text{Consumers Pay}} + \underbrace{-\frac{\Delta Depr}{Revenue_{2000}} - \frac{\Delta Profit}{Revenue_{2000}}}_{\text{Firm Owners Pay}}$$

	(1) Changes between 2000 and 2002	(2) Changes between 2000 and 2004
Change in Total Labor Cost relative to Revenue in 2000	0.038	0.021
Change in Revenue relative to Revenue in 2000 ($\Delta Revenue$)	0.066	0.036
Change in Materials relative to Revenue in 2000 ($\Delta Material$)	0.033	0.014
Change in MiscItems relative to Revenue in 2000 ($\Delta MiscItems$)	0.005	0.005
Incidence on Consumers ($\Delta Revenue - \Delta Material - \Delta MiscItems$)	0.028	0.017
Change in Profits relative to Revenue in 2000 ($\Delta Profit$)	-0.011	-0.008
Change in Depreciation relative to Revenue in 2000 ($\Delta Depr$)	0.001	0.003
Incidence on firm-owners ($-\Delta Profit - \Delta Depr$)	0.010	0.005
Fraction paid by consumers (in percent)	74	77
Fraction paid by firm-owners (in percent)	26	23

Evidence from Hungary



Figure 5: The relationship between household income and the minimum wage content of consumption



Who Pays For the Minimum Wage?



UCL

1. Wage retrenchment of higher-skilled workers **No**
2. Passthrough to firm owners **No/Modest**
3. Passthrough to suppliers **Probably no, but little evidence**
4. Passthrough to consumers **Yes**
5. Improving efficiency (reorganization or better allocation) **Yes**

Effect on Productivity



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Growing literature on efficiency, and other cost reductions

- Quantity change (Q) in certain sectors
 - Agriculture, Ku (2022); Coviello et al. (2021) in retail; exporting firms (Hau, Huang and Wang 2019)
- Quality of service from nursing home (Ruffini, 2021)
 - Consistent with increased prices and limited consumer responses
- Indirect evidence on cost reduction through turnover
 - Turnover falls, and so training costs fall (Portugal and Cardoso, 2006; Brochu and Green, 2013; Dube, Lester, Reich, 2016; Ruffini, 2021)
- Change in efficiency through market allocation (Dustmann et al., 2021)
 - Advantage: consistent with higher prices leading to higher quality

Reallocation and the Minimum Wage



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- Dustmann, Lindner, Schönberg, Umkehrer, and vom Berge (2022) provide direct evidence on the impact of the minimum wage on reallocation
- **Reform:** Introduction of the minimum wage in 2015
- **Data:** administrative employer-employee database covering the life history of all German private sector workers
- Empirical analysis: how wages, employment, and quality of the firm evolved following minimum wage hikes
- “Control” group
 1. What trends do we observe before the introduction of the minimum wage?
 2. How did wages evolve at the higher end of the wage distribution?

Winston Churchill – Trade Boards Act in 1909



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- **Trade Boards Act 1909:** a piece of social legislation passed in the United Kingdom in 1909
- It provided the creation of boards which could set minimum wage criteria that were legally enforceable
- Winston Churchill, MP, put the argument for the legislation as follows:

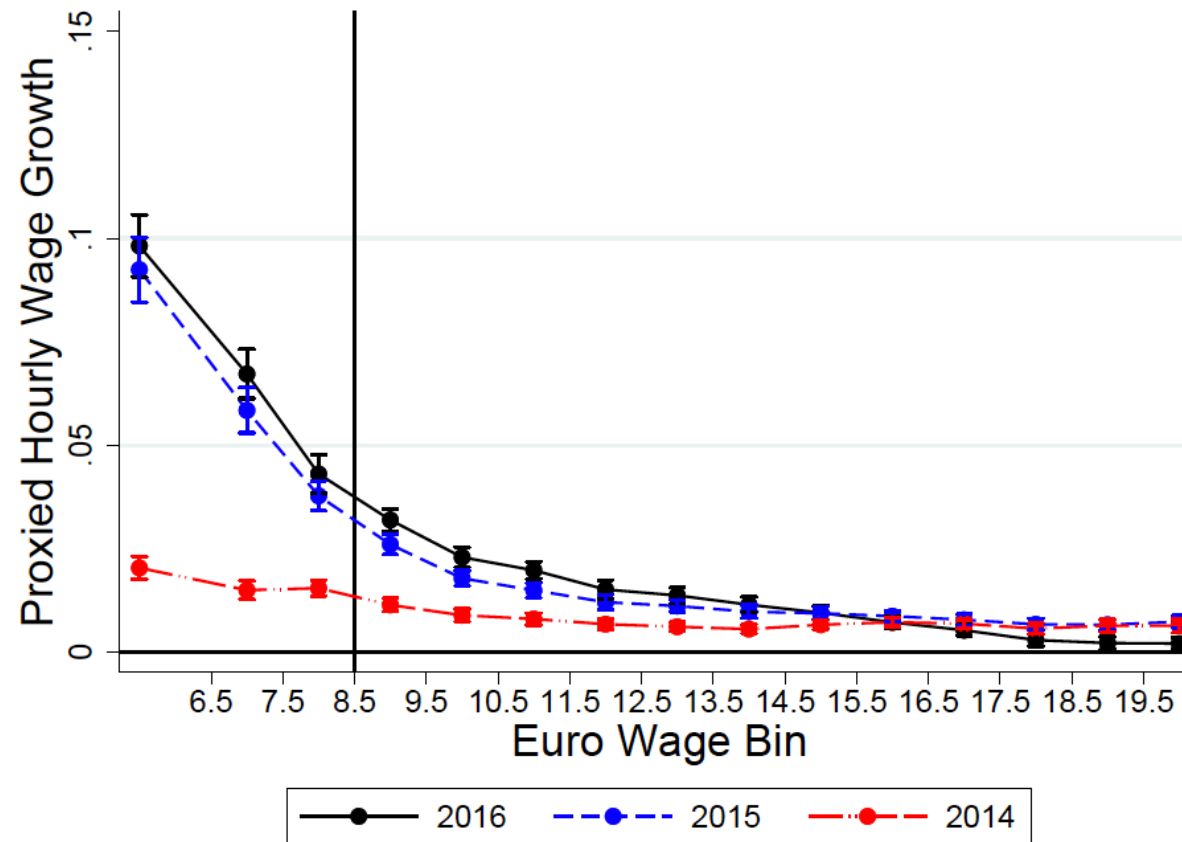
It is a serious national evil that any class of His Majesty's subjects should receive less than a living wage in return for their utmost exertions. It was formerly supposed that the working of the laws of supply and demand would naturally of regulate or eliminate that evil. The first clear division which we make on the question today is between healthy and unhealthy conditions of bargaining. [...]

[...] But where you have what we call sweated trades, you have no organisation, no parity of bargaining, the good employer is undercut by the bad, and the bad employer is undercut by the worst; the worker, whose whole livelihood depends upon the industry, is undersold by the worker who only takes the trade up as a second string [...] where those conditions prevail you have not a condition of progress, but a condition of progressive degeneration.

Wage Growth as a Function of Wages 2 Years Before



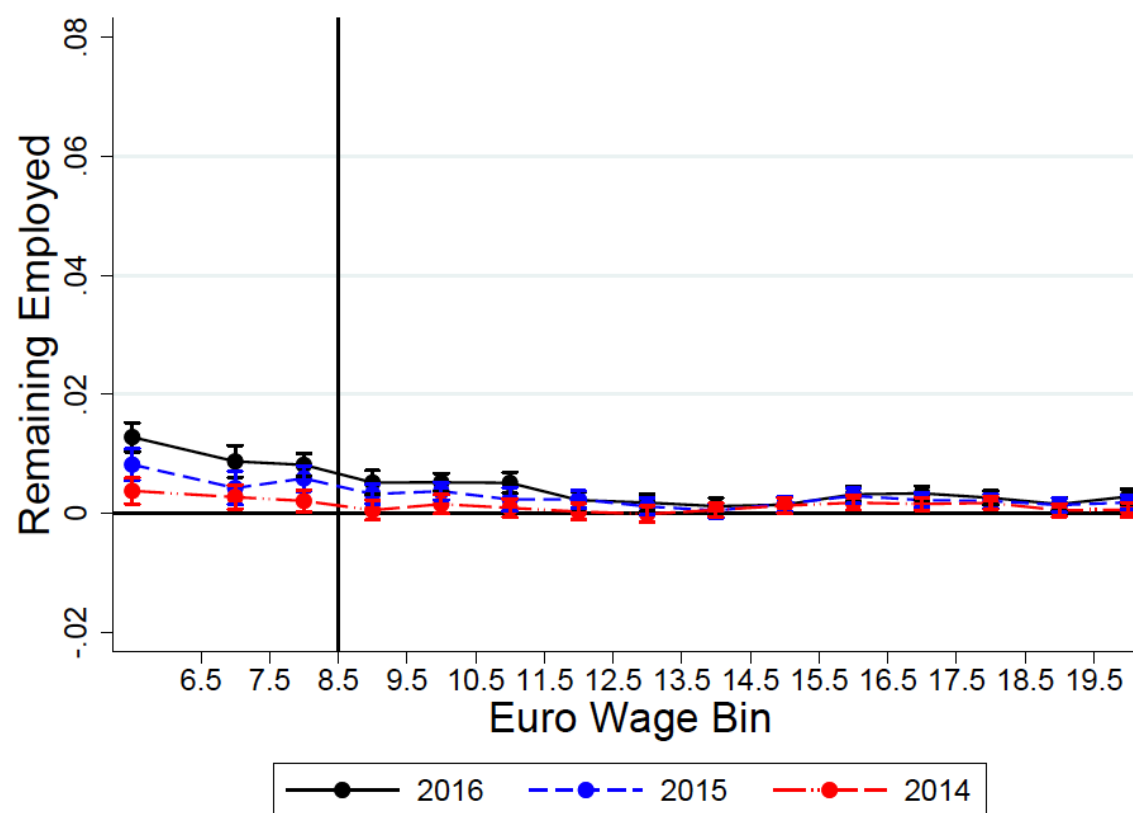
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Employment Growth as a Function of Wages 2 Years Before



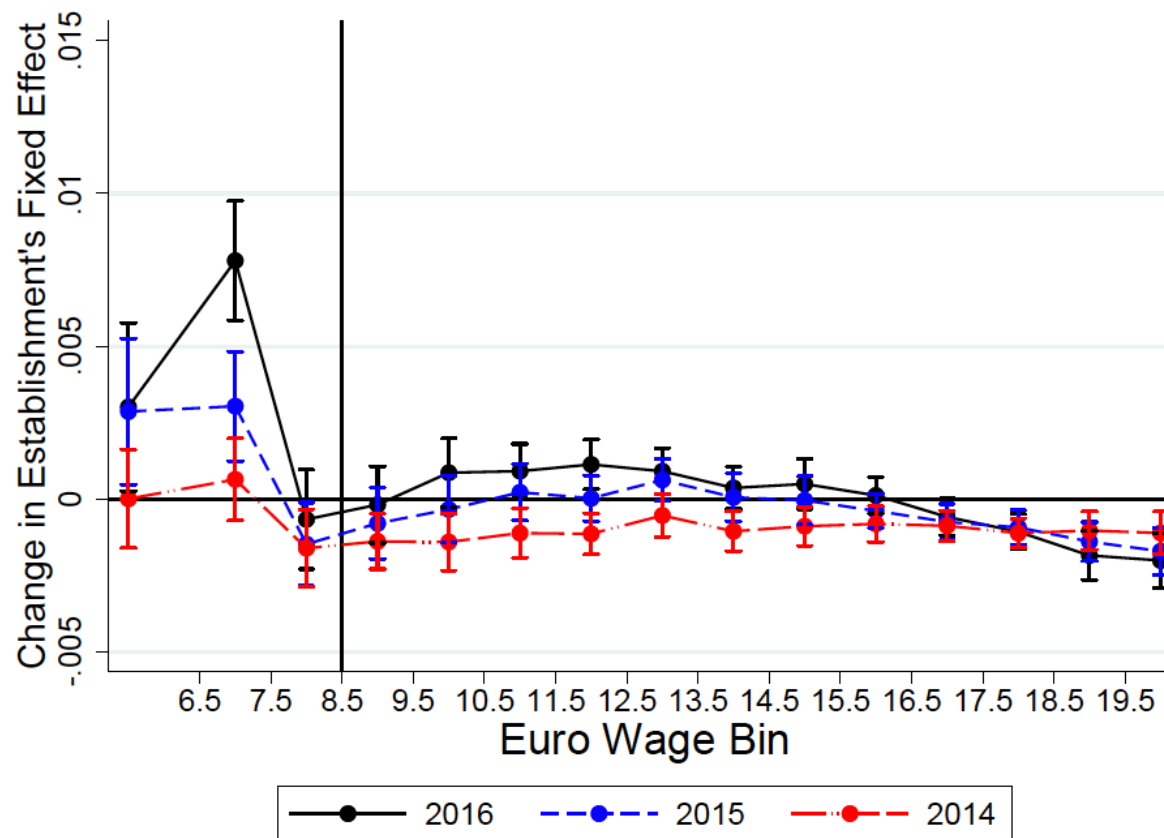
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Firm Wage Premium as a Function of Wages 2 Years Before



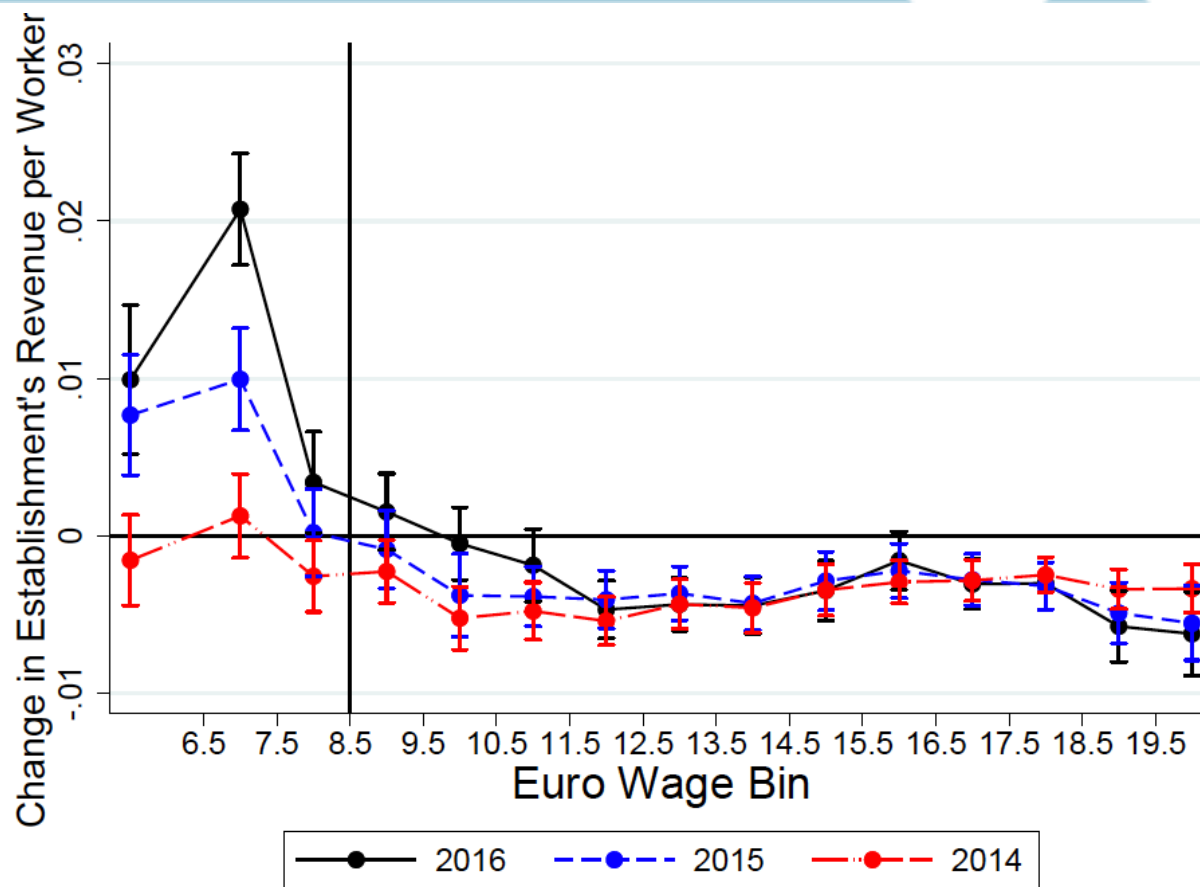
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Firm's Productivity as a Function of Wages 2 Years Before



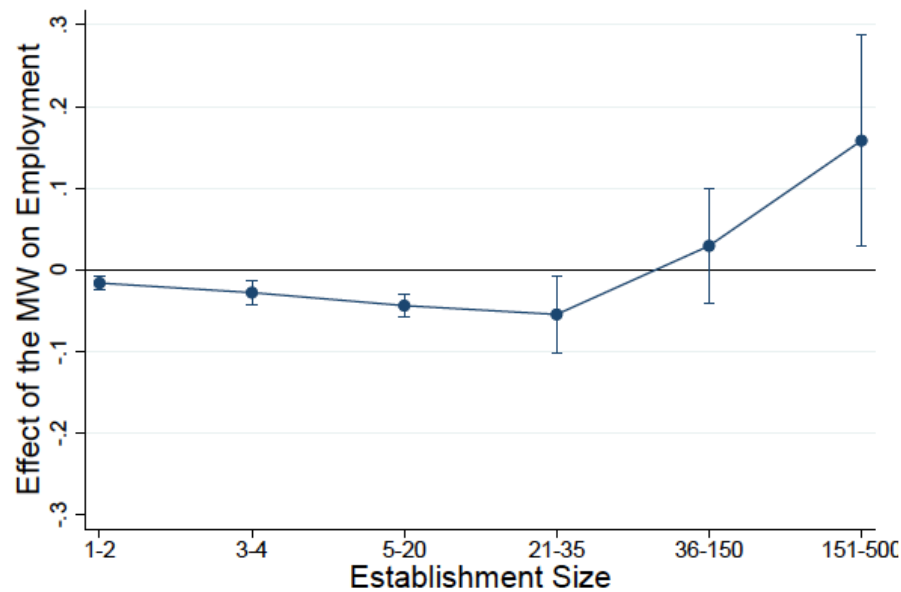
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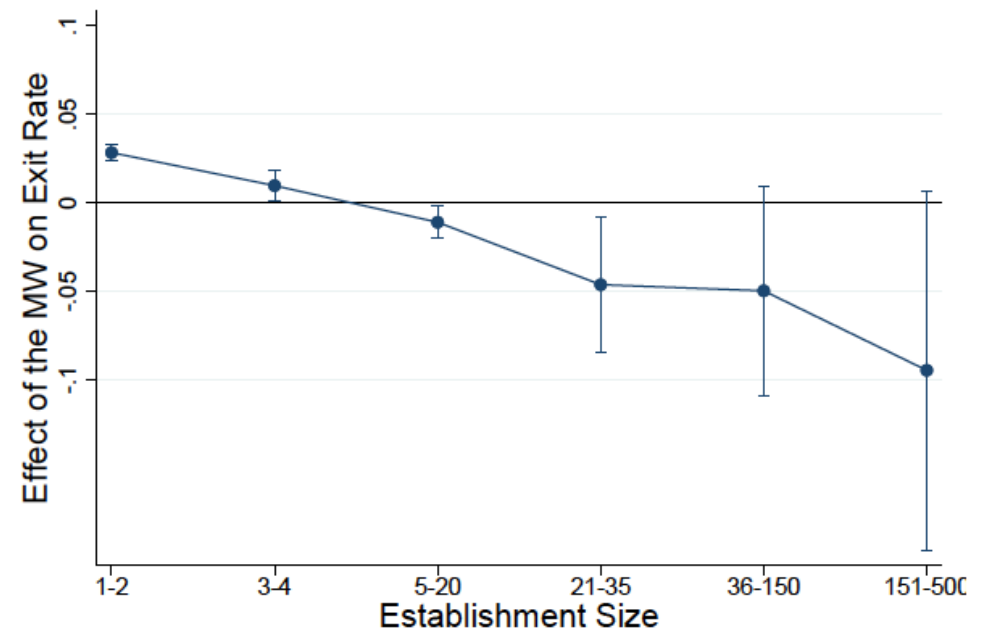
The Effect of the Minimum Wage by Firm Size



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(c) Employment, 2016 vs 2014



Reallocation: Summary of Findings



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Key Findings:

- Positive and significant effect on wages, no disemployment effect
- MW leads to **reallocation of workers** to
 - firms paying higher wage premium
 - firms with lower turnover
 - larger and more productive firms

Further Evidence

- Increases commuting time
- MW at highly exposed locations led to
 - a decrease in the number of firms
 - an increase in average firm size
 - an increase in average firm wage premium and productivity

Reallocation: Efficiency and Welfare



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Efficiency

- MW led to worker reallocation from less to more productive firms
- Productive efficiency of worker-firm sorting was increased

Welfare

- Workers' commuting distance was also increased
- The effect on worker's welfare is unclear, but our model suggests that it was increased on average

Monopsonistic competition model Revisit



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- This reallocation naturally emerge in our monopsonistic competition model with heterogenous firms

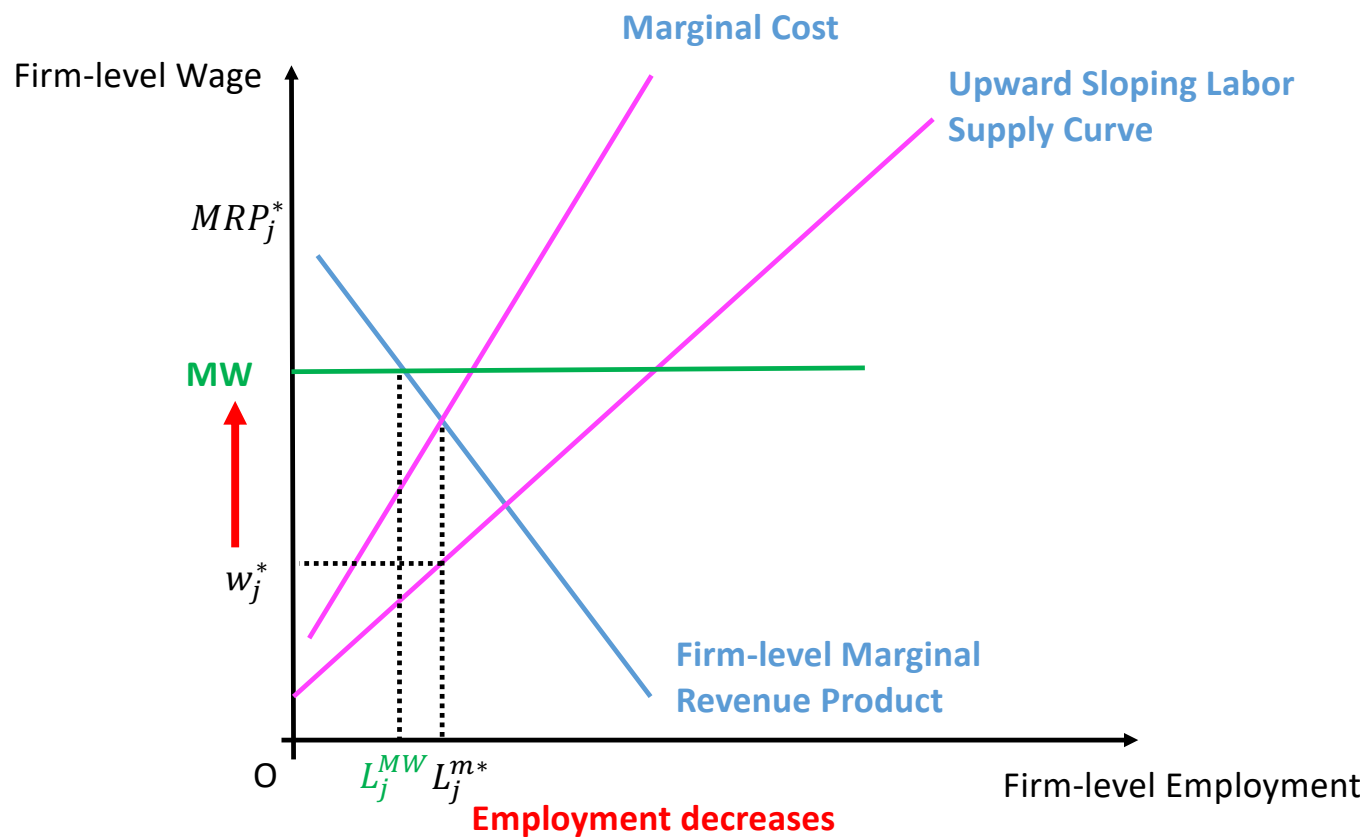
Three regimes:

- Low productive firms
- Medium productive firms
- High productive firms

Low Productive (Demand constrained)



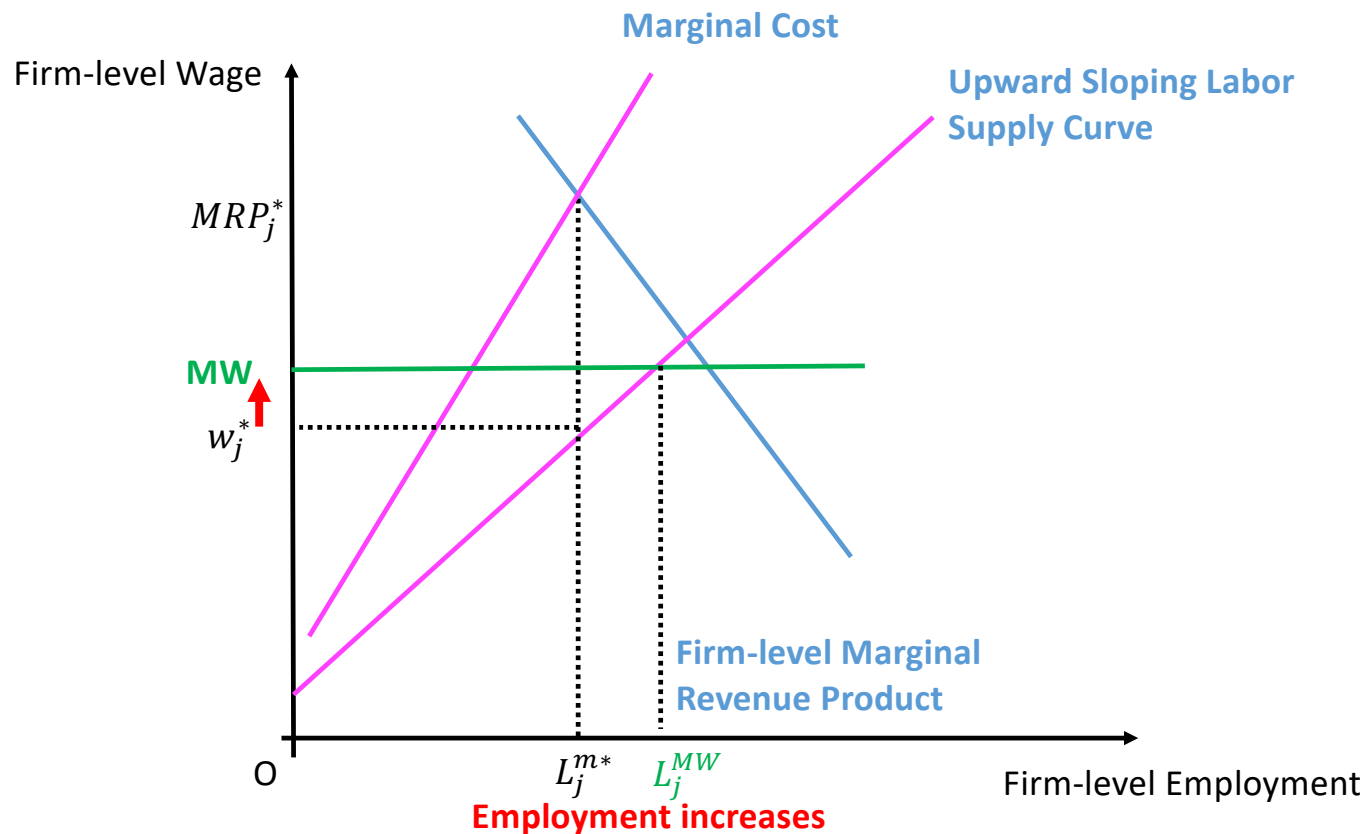
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Low Productive (Supply constrained)



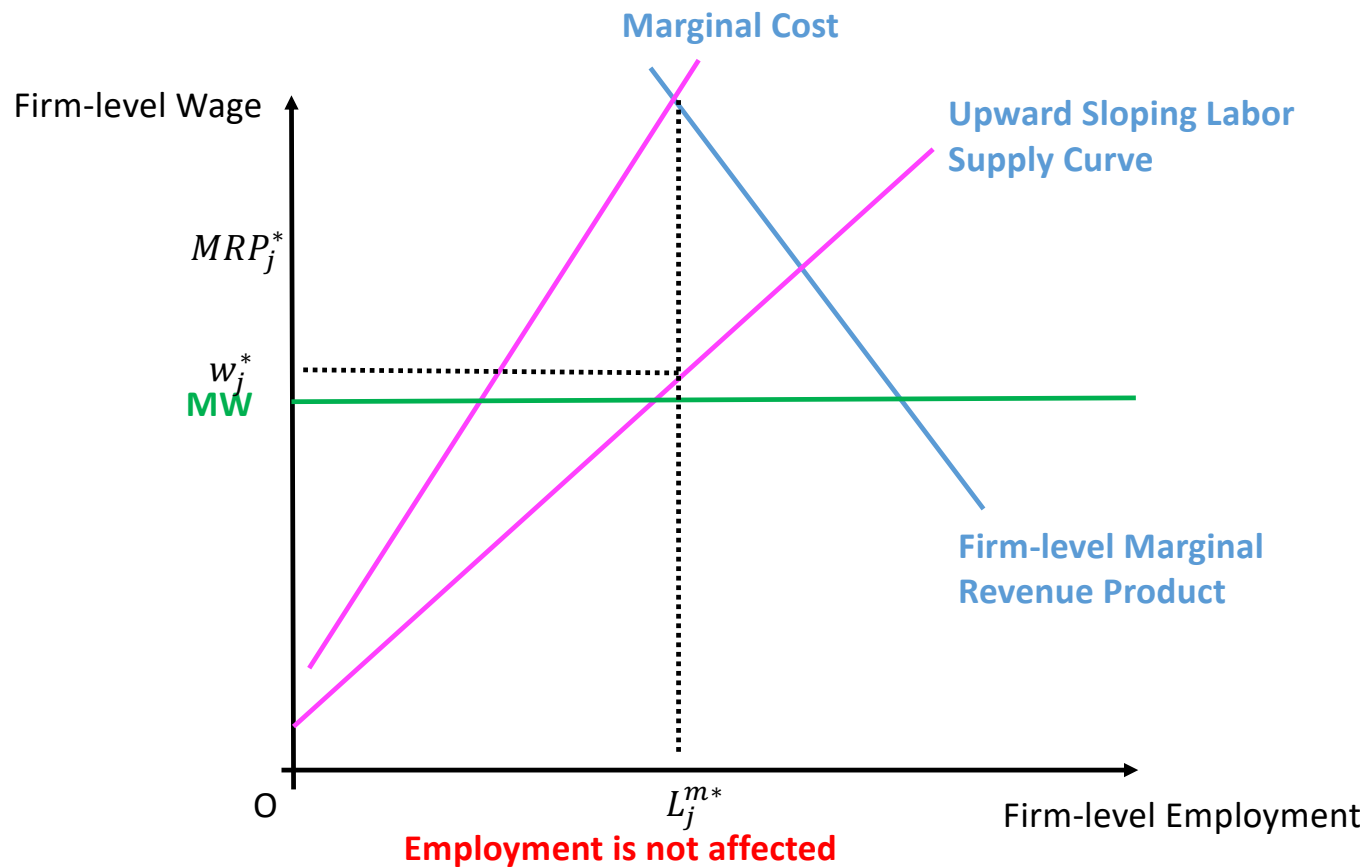
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High Productive (Not binding)



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The Model of Workplace Differentiation



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What is the impact of minimum wages in this framework?

Three regimes:

- Low productive firms (destroy jobs)
- Medium productive firms (creates jobs)
- High productive firms (no effect)

Reallocation from less efficient to more efficient firms



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Summary of the review

Review of the Evidence



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Our conclusion:

- Most important channel: *pass-through to consumers*
- Improvement in efficiency plays a role
 - Reduction of other costs (e.g. lower training cost from reduced turnover)
 - Improved efficiency (probably through improved quality)
- Likely only a small reduction in profits (but limited and somewhat mixed evidence)

What manager survey says?



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Business surveys (Lester, 1946, Bodnar et al., 2018, Reich and Laitinen 2003 WP)

- Limited employment response
- Increase sales effort and output prices
- Improved management and efficiency

Very much consistent with what the evidence accumulated on various adjustment margins in the past 30 years!

Question to contemplate about?

- Worth asking managers about how do they react to various shocks

Detour



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Puzzle



- Standard competitive model predicts that MW is passed through to consumers via higher prices:

Higher labor cost $\rightarrow$ Prices increase $\rightarrow$ Output demand falls $\rightarrow$ Emp. Falls

- Monopsony predicts employment increases

Emp. Increases $\rightarrow$ Firms sells more $\rightarrow$ Prices fall

Empirically, we see positive price effects and a limited employment change

Potential Solutions to This Puzzle



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- Ignore evidence on employment or on prices
 - Typically employment responses are ignored even though prices are much harder to measure (see e.g. French and Aaronson, 2007)
- MW-induced price changes might be different? (coming soon Becker et al.)
 - Consumers are willing to pay more for goods and services produced under fair working standards (Hiscox and Smyth, 2020)
 - Some firms cannot credibly commit to higher prices and higher wage standards
 - MW solves this credibility problem by raising wage standards and prices at the same time

Potential Solutions to This Puzzle



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- Assess whether consumer responses to price changes depend on the “narrative” around the price change
 - Anecdotally, firms provide justifications for prices
 - Consumer demand elasticity might depend on reason provided and whether it is credible (believed)
 - Impact of various government policies depends on consumers’ reactions to price changes
- Survey Experiments
 - Present survey participants with hypothetical scenarios describing a price increase
 - Randomize the reason for price increase and product being purchased

Treatments: Reasons for Price Increase



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1. No reason: **Control**
2. Minimum wage: **Main treatment**
3. Rent: **Justified price increase**
4. Energy prices: **Justified increase affecting all shops**
5. Taxes: **More general redistributive externality**
6. Voluntary minimum wage: **Verifiability**

Treatment 1 - Control Treatment



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A local coffee shop in your neighborhood has increased its prices by 25%. This means that if a coffee cost \$3 before the change, it now costs \$3.75.

Assuming you had visited this coffee shop before, do you think you would be more or less willing to purchase items from the coffee shop following the price increase?

- ☐ More willing
- ☐ As willing as before
- ☐ Less willing

Treatment 2 – Minimum Wages



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A coffee shop in your neighborhood spends roughly half of its total revenue on employee wages. Your state government has announced that it will raise its minimum wage by 50%, meaning that the coffee shop will have to pay many of its employees a higher hourly wage. The owner of the coffee shop therefore decides to increase prices by 25%. This means that if a coffee cost \$3 before the change, it now costs \$3.75.

Assuming you had visited this coffee shop before, do you think you would be more or less willing to purchase items from the coffee shop following the price increase?

- Ⓐ More willing
- Ⓑ As willing as before
- Ⓒ Less willing

Treatment 3 – Energy



Businesses in your area have been affected by energy prices, which have risen by 250%. The owner of a coffee shop in your neighborhood spends 10% of his revenue on energy and has therefore decided to increase prices by 25%. This means that if a coffee cost \$3 before, it now costs \$3.75.

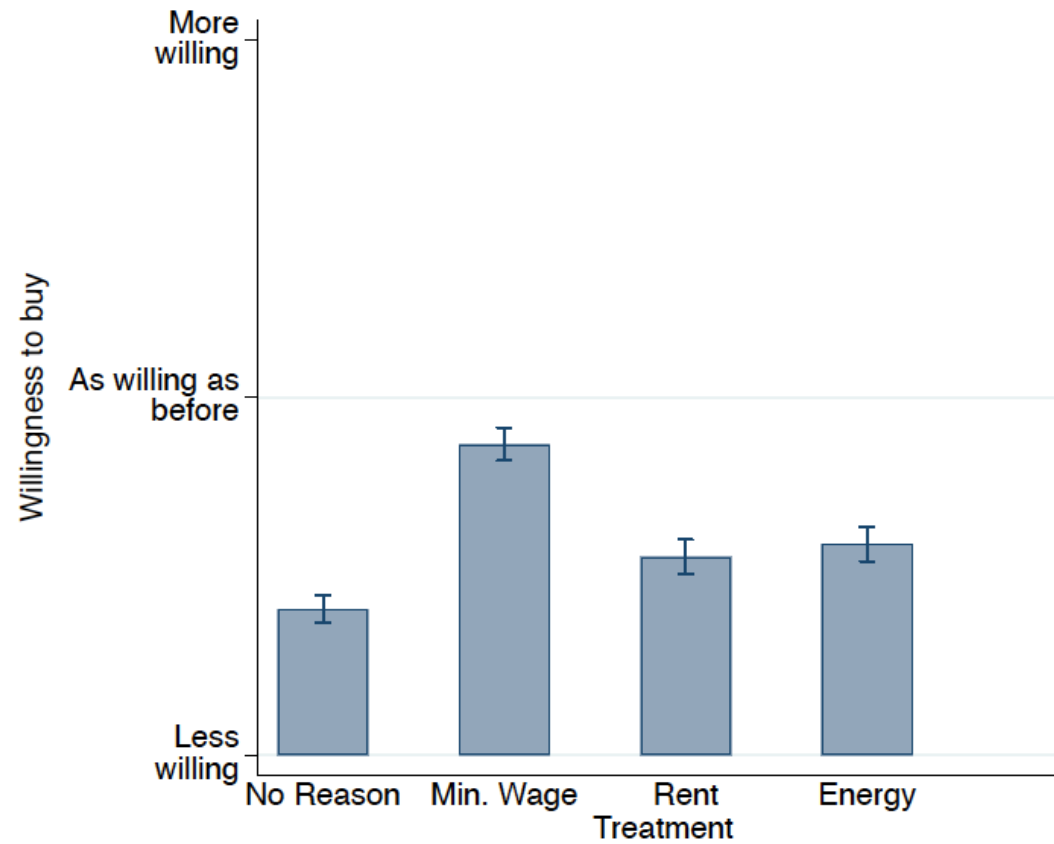
Assuming you had visited this coffee shop before, do you think you would be more or less willing to purchase items from the coffee shop following the price increase?

- Ⓐ More willing
- Ⓑ As willing as before
- Ⓒ Less willing

Main Results



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Effects across Different Price Changes



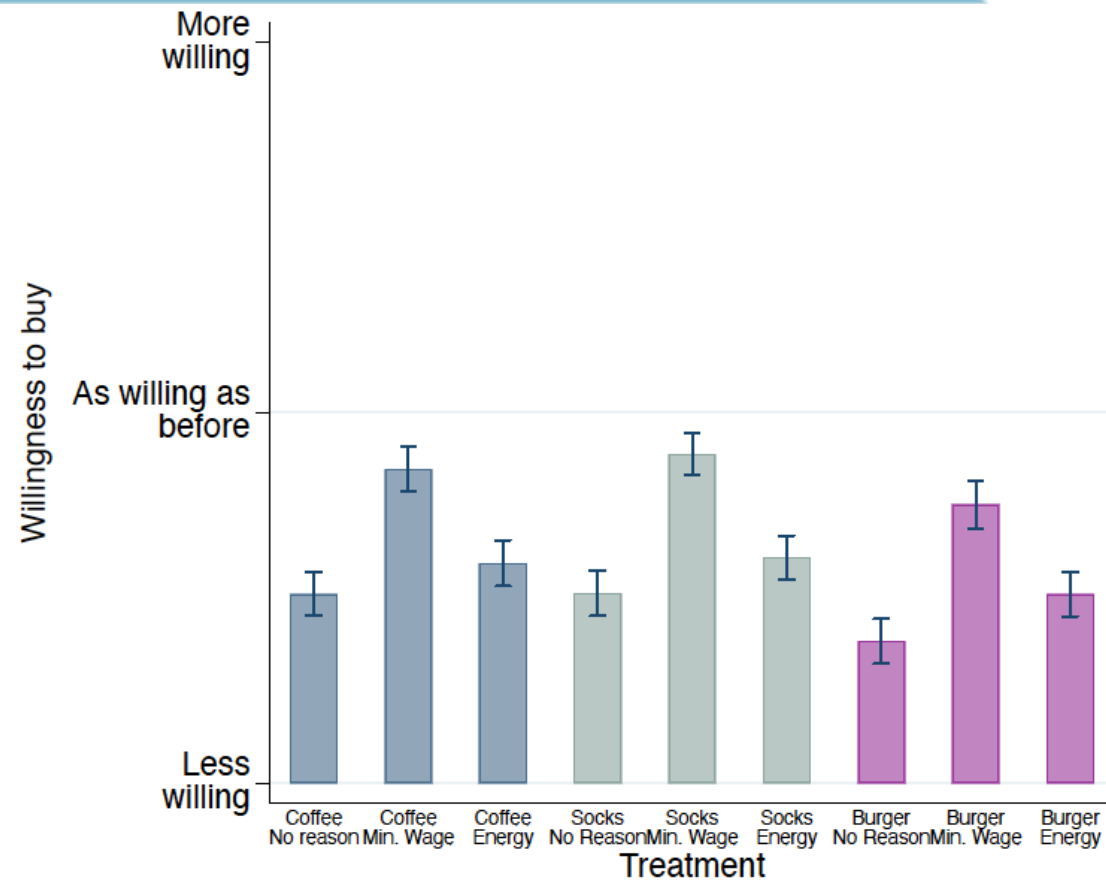
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Effects across Different Products



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Effects across Different Products



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- ▶ Whereas the presented scenarios were intended to reflect a situation with 100% pass-through, participants in our study might have had differing beliefs regarding the pass-through
- ▶ In the last wave of the experiment we therefore asked directly about pass-through (in treatments on minimum wage and energy prices):

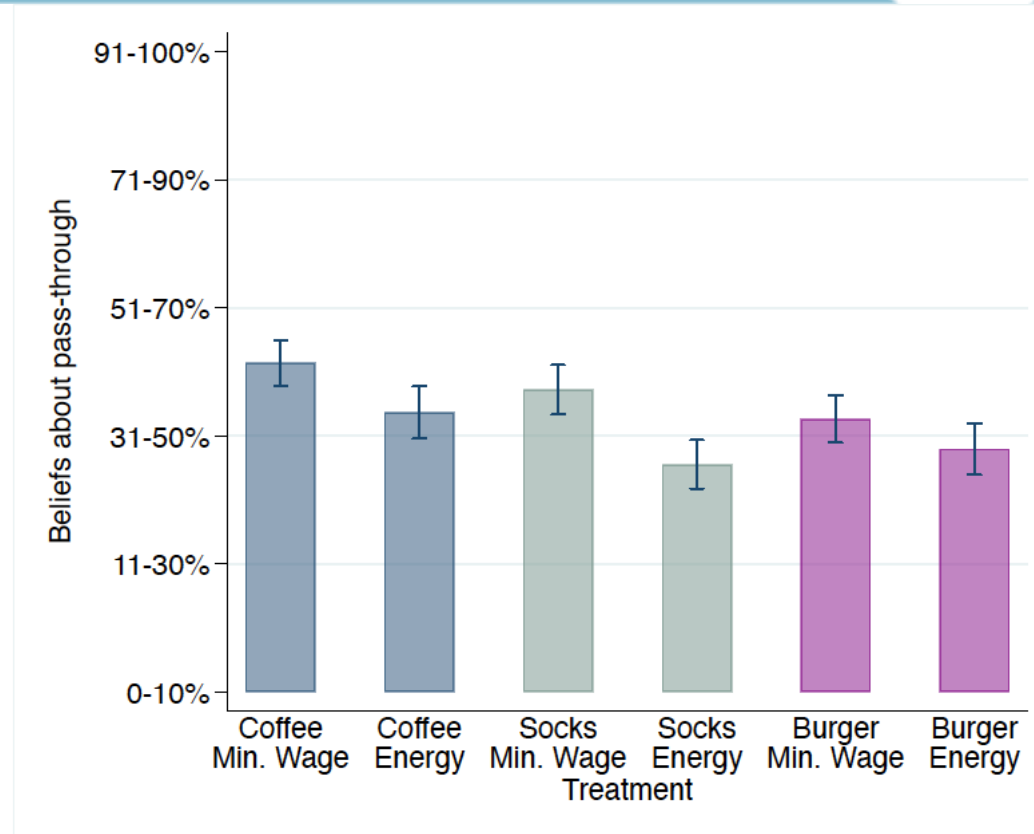
What percent of the price increase do you believe goes toward the workers' wages? 100% means that all of the price increase goes to workers' wages and 0% means none of the price increase goes to workers' wages.

- Ⓐ 0-10%
- Ⓐ 11-30%
- Ⓐ 31-50%
- Ⓐ 51-70%
- Ⓐ 71-90%
- Ⓐ 91-100%

Effects across Different Products



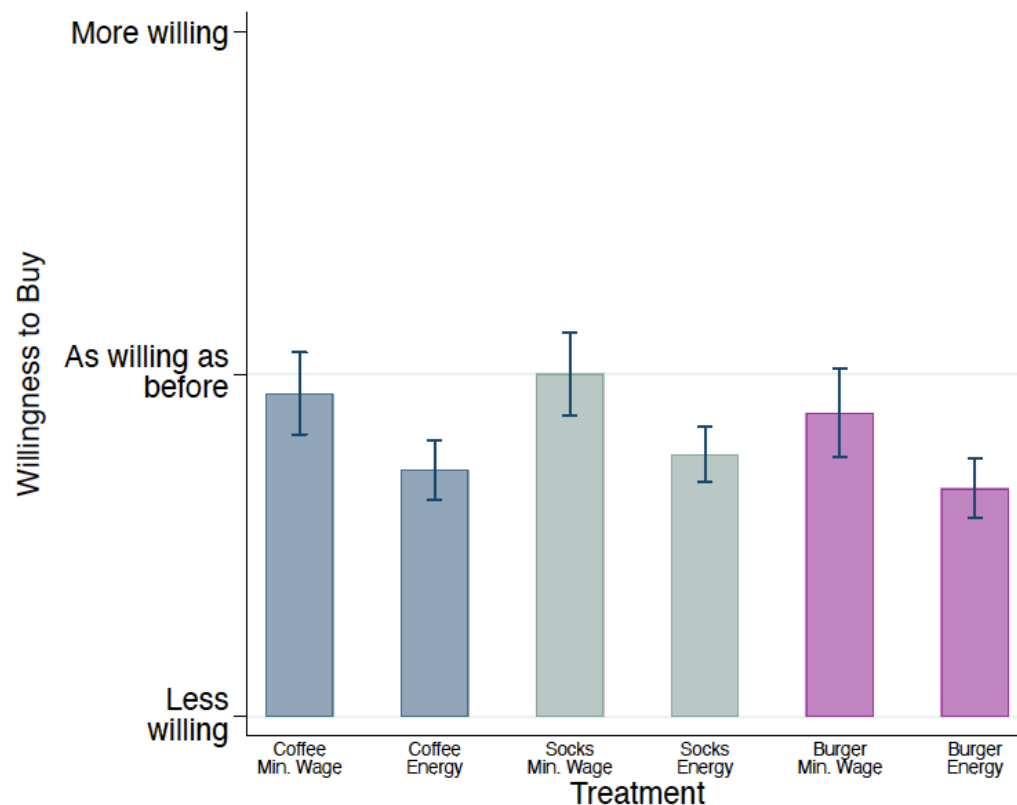
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WTB conditional on Beliefs regarding Pass-Through



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Tax Treatments

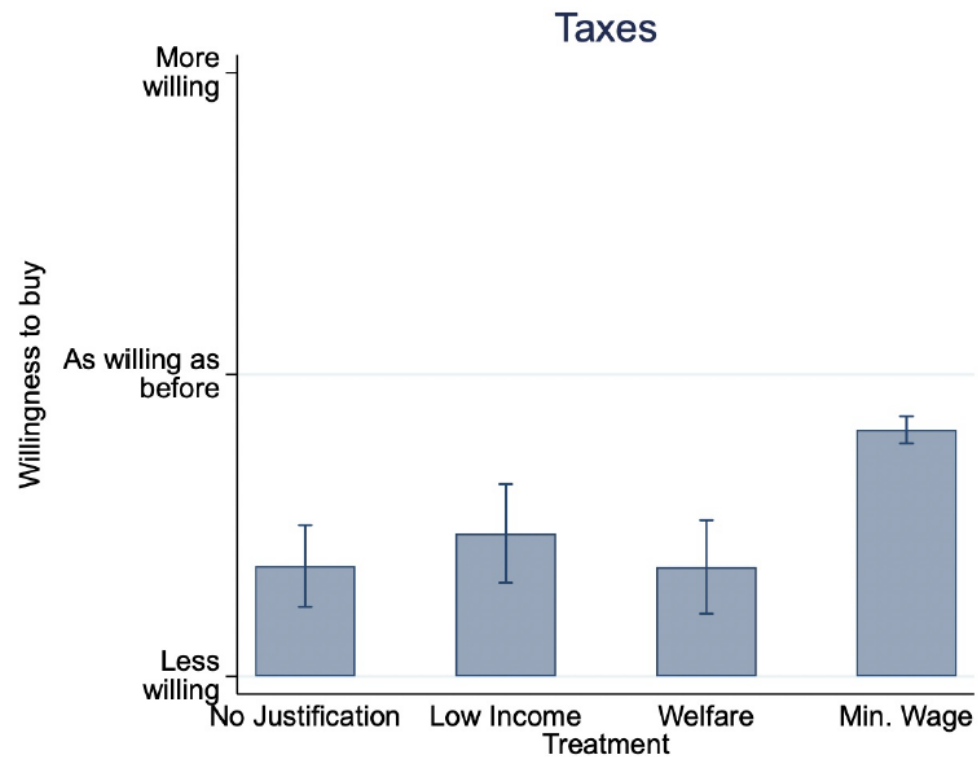


- More general (narrowly bracketed) social preference over the redistributive consequences of one's consumption?
- Consumption tax increase:
 - No justification
 - Extra revenue to be spent to increase the generosity of welfare programs
 - Extra revenue to be spent on tax cuts for low-income people

Tax Treatments



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Welfare Implications for the MW



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The reasons behind inelastic demand matter for welfare

Standard assumption:

- Demand is inelastic because no good/easy substitute is available for minimum wage-produced goods (e.g. MaCurdy, 2015).
- Implication: demand should be the same for the control and the minimum wage treatment
- Minimum wage induced output price increase reduces the welfare of consumers

Transactional preferences:

- Output demand is inelastic because consumers want objects that are produced “ethically” using higher labour standards
- Minimum wage induced output price increase may not decrease but instead increase the welfare of consumers
- Our survey evidence points to such an explanation

Conclusions



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- MW has limited employment effects at the current levels
- Affects worker-firm allocation, and could potentially increase productivity in certain cases
- Prominent interest in MW, since it allows “testing” competing explanations on how the low-wage labor market operates
- Standard competitive model fails to capture key aspects
 - Note: such model is widely used to study the impact of taxes and transfers on low-wage workers